

GASTRO-INTESTINAL SURGERY

Gastro-Esophageal Reflux Disease (GERD)

Introduction

- Reflux esophagitis is inflammation of the lower esophageal mucosa ^{2^{ry}} to reflux of gastric and/or duodenal contents up into the esophagus.
- Gastro-esophageal reflux may be asymptomatic and is regarded as *physiological*. Short periods of reflux, especially after eating can also be regarded as normal.
- *Significant* reflux, leading to symptoms (including nocturnal episodes), is termed gastro-esophageal reflux disease (GERD), which is now a common cause of dyspepsia in the 'developed' world and can lead to esophagitis and its complications.

Epidemiology

- GERD affects up to 60% of the population at some time in their lives, although up to 40% of these patients will have no evidence of esophagitis at endoscopy.
- It is rare in certain parts of the world (e.g. the Orient) but is increasing rapidly in the *developed* world and represents a major drain on health care resources. Not only is the cost of drug therapy and surgery for GERD increasing dramatically, but GERD also causes significant economic consequences in terms of work absenteeism.

Factors Involved in the Normal Prevention of Reflux

1. Lower esophageal sphincter (LES): (a) Intrinsic LES pressure, (b) Total length of LES and (c) intra-abdominal length of LES
2. Angle of His ('flap valve' mechanism)
3. Crural fibers of the diaphragm ('pinchcock' action)
4. Mucosal rosette (prominent mucosal folds at the gastro-esophageal junction - GEJ)
5. Swallowed saliva (lubrication and neutralization)
6. Ante-grade esophageal peristalsis
7. Normal gastric motility/emptying.

Pathophysiology

- There is an underlying anatomical or physiological defect leading to failure of the normal anti-reflux mechanism. The major cause of significant GERD is inappropriate transient LES relaxations (TLESRs).

- Substances that have been implicated include smoking, alcohol, coffee, chocolate and excess dietary fat intake. The role of *Helicobacter pylori* (*H. pylori*) is *uncertain*.

Causes of GERD (Table 1)

Table 1: Primary and secondary causes of gastro-esophageal reflux disease (GERD)

Primary Causes	Secondary Causes
1. Decreased LES pressure 2. Decreased total LES length (<2–3 cm) 3. Decreased intra-abdominal LES length (<1–2 cm) 4. Loss of angle of His 5. Sliding hiatus hernia 6. Diaphragmatic crural defect 7. Loss of mucosal rosette (e.g. due to inflammation)	1. Poor esophageal motility 2. Excess gastric acid production 3. Impaired gastric emptying 4. Excess duodeno-gastric reflux 5. Decreased saliva production

Clinical Features of GERD

- *Typical symptoms* include heartburn, retrosternal or epigastric pain, regurgitation and water brash, dysphagia and odynophagia (painful swallowing).
- *Atypical symptoms* include back pain, cardiac-type chest pain, chronic wheeze or asthma (especially at night), sore throat, hoarse voice, halitosis or dental decay.

Investigations

Endoscopic Examination

- It is the mainstay of investigation for GERD, allowing direct visualization of the lining of the esophagus and stomach.
- Esophageal abnormalities that can be diagnosed at endoscopy include hiatus hernia (HH), esophagitis, Barrett's esophagus, strictures and tumors.
- The severity of reflux esophagitis can also be graded at endoscopy, and *biopsies* can be taken from the mucosa to facilitate histological examination of the tissues.
- Other diagnostic techniques employed at endoscopy include brush cytology and washings, vital staining and fluorescence detection to ↑ recognition of minor mucosal abnormalities.

Radiological Contrast Studies

- *Barium swallow* may show structural abnormalities (e.g. strictures, tumors and diverticula). It also yields information regarding esophageal function and the competence of the LES.
- *Video fluoroscopy* allows dynamic visualization of esophageal function and can demonstrate reflux of contrast up into the esophagus.
- Although radiological contrast studies have been largely superseded by endoscopy, barium swallow and video fluoroscopy still have a role, especially where esophageal function needs to be demonstrated and in patients who refuse endoscopy.

Esophageal pH Measurement

- Intra-esophageal pH measurement is performed as an ambulatory test over a 24-hour period. This involves the passage of a fine pH probe into the lower esophagus, which is connected to a small recording device worn by the patient.
- This makes it possible to measure the duration, frequency and severity of reflux attacks, and it is now the ‘gold standard’ means of quantifying the degree of acid reflux.

Esophageal Manometry

- Using this technique, it is possible to determine the length and position of the LES and to measure the LES pressure.
- A short or weak LES is often a major cause of reflux esophagitis. A "LES" pressure of < 4 mmHg is abnormal and may be an indication for surgery. It is also possible to measure the resting and maximum contracting pressures at various levels within the esophagus and to study the propagation of peristaltic waves down the esophagus.

Classification of Reflux Esophagitis (Endoscopic Examination)

Savary-Miller Classification

This was devised in 1967 by Savary and Miller, who graded the degree of esophagitis into grades 1 to 4, based on the extent of the inflammation and/or erosions, together with the presence or absence of complications, such as scarring, strictures or columnar metaplasia as shown in Table 2.

Table 2: Savary-Miller classification (grading) of reflux esophagitis

Grade	Description
<i>Grade 1</i>	Erythema or diffusely red mucosa; edema causing accentuated fluid loss
<i>Grade 2</i>	Isolated round or linear erosions extending from the gastro-esophageal junction in relation to the folds
<i>Grade 3</i>	Confluent erosions extending around the entire circumference or superficial ulceration, but without stenosis
<i>Grade 4</i>	Complicated: erosions (as described above) plus deep ulceration, stricture or columnar-lined esophagus

Los Angeles Classification

The Los Angeles Classification represents a further attempt to subdivide uncomplicated reflux esophagitis into 4 grades based on both the length and width of the lesions as shown in Table 3.

Table 3: Los Angeles Classification (grading) of reflux esophagitis

Grade	Description
<i>Grade A</i>	One (or more) mucosal break(s) no longer than 5 mm that does not extend between the tops of 2 mucosal folds
<i>Grade B</i>	One (or more) mucosal break(s) longer than 5 mm that does not extend between the tops of 2 mucosal folds
<i>Grade C</i>	One (or more) mucosal break(s) that is continuous between the tops of 2 or more mucosal folds, but which involves < 75% of the esophageal circumference
<i>Grade D</i>	One (or more) mucosal break(s), which involves at least 75% of the esophageal circumference.

Complications

The 2 most common complications are benign stricture and Barrett's esophagus.

1. Benign (Peptic) Strictures

- These occur in inadequately treated patients.
- They present with dysphagia and endoscopy is essential to exclude malignancy.
- Treatment is by endoscopic dilatation followed by either long-term proton-pump inhibitors (PPI) or anti-reflux surgery to prevent recurrence.

2. Barrett's Esophagus

- It is the replacement of normal squamous epithelium by metaplastic columnar epithelium due to long-standing reflux. Its importance lies in its malignant potential.
- In order to diagnose Barrett's esophagus, intestinal metaplasia must be identified histologically.
- Treatment is a combination of acid suppression and life-long endoscopic surveillance. Anti-reflux surgery may have a role and endoscopic ablation techniques have produced encouraging early results.

Hiatal (Hiatus) Hernia (HH)

Definition

- It is the "protrusion (or herniation) of the upper part of the stomach into the chest cavity through the esophageal hiatus because of a tear or weakness in the diaphragm".

Classification

- Four types of esophageal hiatal hernia are identified (Table 4) (Figure 1)

Type of HH	Description
Type I HH (Sliding) (85%)	It is the cephalad displacement of the gastro-esophageal junction (GEJ) through the hiatus into the mediastinum. It is usually small, asymptomatic and reducible. Commonly associated with GERD.
Type II HH (Rolling) (10%) = para-esophageal	It is superior migration of the fundus of the stomach alongside the GEJ and esophagus into the mediastinum with GEJ in normal intra-abdominal location. May present by abdominal and chest pain, hiccough, early satiety, regurgitation, post-prandial bloating, arrhythmia, dysphagia & dyspnoea. About 40% of patients present with acute features of perforation/gangrene/ bleeding.
Type III HH	It is combination of type I and type II
Type IV HH	HH containing other abdominal viscera e.g. transverse colon and omentum

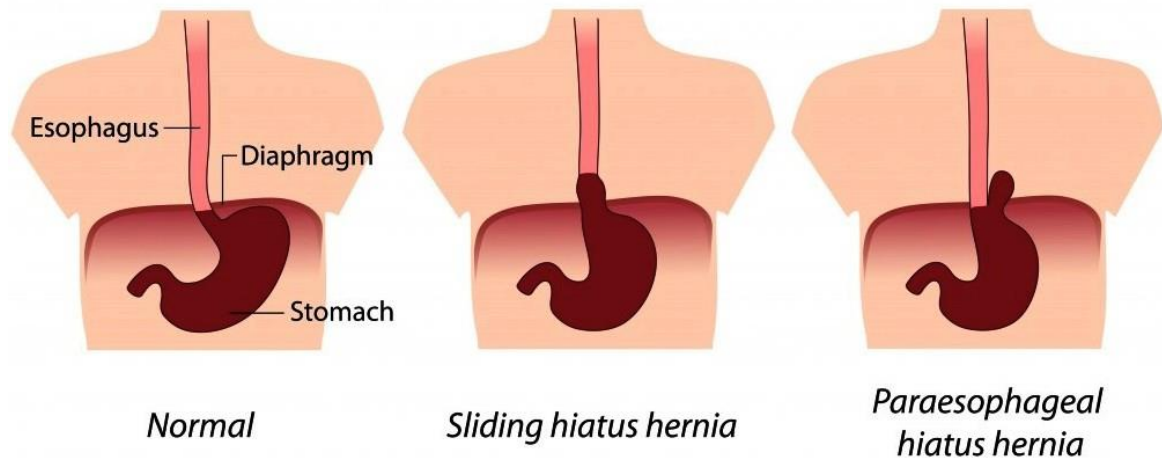


Figure 1. Hiatus hernia: sliding (Type 1) & para-esophageal (rolling) (Type 2)

Management

General Measures

- Avoidance of certain foods and drinks (alcohol, coffee, chocolate and fatty or spicy foods).
- Diet (weight loss).
- Cessation of smoking (because smoking has a potent effect at lowering LES pressure)
- Other lifestyle changes include avoidance of tight abdominal garments, alteration of eating habits (change in timing, type and volume of meals) + elevation of the head of the bed.

Medical Therapy

- *Antacids* act by neutralizing the acid pH within the stomach & lower esophagus, but their benefits are often short-lived.
- *H₂-receptor antagonists* (e.g. Cimetidine, Ranitidine, Famotidine, Nizatidine) have been largely superseded by PPIs, which have a more powerful acid-suppressant effect.
- *Proton-pump inhibitors (PPIs)* (e.g. Omeprazole, Lansoprazole, Pantoprazole, Rabeprazole) are powerful acid suppressants and now form the mainstay of Rx for GERD.
 - They block the proton pump within the mucosal cells of the stomach lining, thus inhibiting the active transport of hydrogen ions across the membrane and thereby almost completely abolishing gastric acid production. Their clinical effect better than H₂-receptor antagonists, leading to better acid suppression and more effective symptom control and healing of ulcerative esophagitis in most patients. A full-dose therapy should be started until the esophagitis has healed, followed by long-term maintenance therapy using a reduced (usually half-strength) dose.
 - Recently, a more powerful ‘second generation’ PPI has also been introduced. *Esomeprazole* is the s-isomer of Omeprazole & has even greater & more long-lasting anti-secretory activity. At present, it is largely used in more severe disease (grades III and IV) or as 2nd-line treatment where ‘first generation’ PPIs failed to control symptoms.

- *Mucosal protection agents* e.g. Sucralfate help to protect against other (non-acidic) agents in the refluxate. However, sucralfate has an unpleasant taste and may produce constipation.
- *Prokinetic agents* e.g. Metoclopramide, Domperidone and Erythromycin, improve gastric emptying and clearance of contents from lower esophagus, but are of little benefit in the treatment of reflux esophagitis
- *Eradication of *H. pylori* infection*: There is little evidence that this is of any benefit in the treatment of reflux esophagitis & is *not* currently recommended as standard treatment.

Surgical Treatment

Advantages

- Anti-reflux surgery is the most effective long-term treatment for GERD as it prevents the reflux of *all* gastric contents.
- Surgery also avoids the problem of poor patient compliance with long-term medication.

Side-effects of surgery:

- Dysphagia (usually short-lived), inability to belch, the so-called ‘gas bloat’ syndrome and excessive flatus.

Indications

1. *Failure of medical treatment* (persistent symptoms or complications despite full-dose PPIs).
2. *Complications* e.g. ulceration, stricturing, Barrett’s esophagus and respiratory problems, such as wheezing or coughing.
3. *Age*: Many patients, especially the young, prefer surgery to prolonged lifelong medication.

Pre-operative assessment

1. Endoscopic evaluation to assess the degree of esophagitis, note the size of HH if present and, more importantly, to exclude other UGI pathology that may mimic symptoms of GERD
2. Esophageal manometry to assess position and function of LES, contractility of the esophageal body musculature and propagation of peristalsis. It is therefore important to exclude achalasia and other esophageal motility disorders. In some centers, the type of anti-reflux surgery is tailored according to the manometric assessment.

Types of anti-reflux surgery

Many types of anti-reflux surgery have been described and most lead to excellent results. Trans-abdominal and trans-thoracic routes have been widely adopted, and both can be performed through either open or minimal access (laparoscopic/thoracoscopic) techniques, (Figure 2).

1. Nissen (360 °) fundoplication:

It is the ‘gold standard’ trans-abdominal operation, in which the fundus of the stomach is wrapped completely around the lower esophagus, following a posterior hiatal repair. The Nissen procedure is very effective at controlling reflux, but it is easy to make the wrap too tight and thus produce dysphagia and gas bloat

2. *Partial fundoplication:*

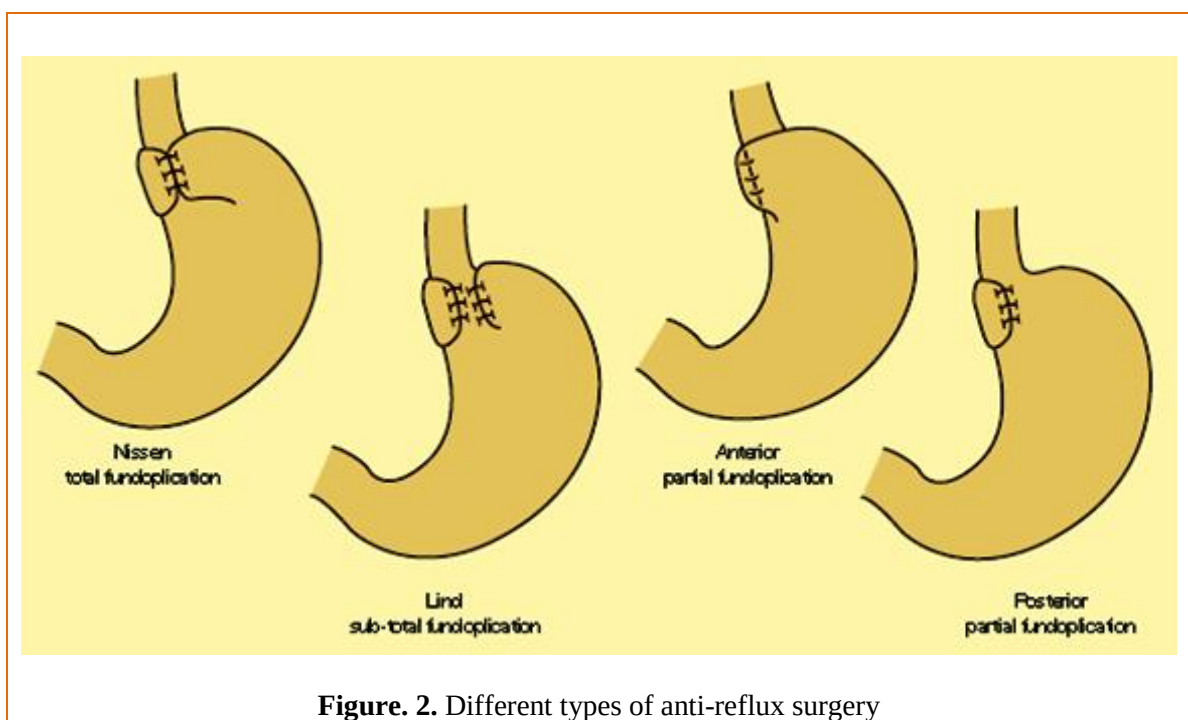
In an attempt to ↓ dysphagia, various partial fundoplications have been described. The rationale behind these procedures is to create a sufficient wrap to prevent reflux but *not* so tight. The major modifications in use are the *Lind* (270°), anterior partial (e.g. *Watson*), posterior partial (e.g. *Toupet*) and sub-total fundoplications. Although the use of a partial fundoplication may ↓ the risk of dysphagia, concerns have been expressed over the long-term durability of these procedures.

3. *Other less common trans-abdominal approaches:*

They include the *Hill* posterior gastropexy, and *Collis*–gastroplasty. The latter is particularly useful in patients in whom the gastro-esophageal junction (GEJ) cannot be reduced below the diaphragm.

4. *Trans-thoracic approach:*

The most common is the *Belsey Mark IV* procedure. Here the lower esophagus is mobilized up to the level of the aortic arch and is then sutured to the gastric fundus and to the diaphragm, to ↓ any hiatus hernia & wrap the fundus over the front of the esophagus.



Results (outcome) of surgery

- Anti-reflux surgery produces effective symptom control in 85-90% of patients.
- The major side-effects include persistent heartburn (5-8%), dysphagia (3-8%), gas bloat and inability to belch and/or vomit.
- Other less common complications include esophageal perforation, pneumothorax, chest infection, splenic damage and deep vein thrombosis (DVT).

Complications of Peptic Ulcer Disease

HEMORRHAGE

Peptic ulceration is responsible for about 44% of all upper gastro-intestinal (GI) hemorrhage. One-third of patients with peptic ulcer (PU) hemorrhage have no previous history or diagnosis of PU disease. Hemorrhage is the leading cause of death due to PU disease with 5-10%. Bleeding usually occurs from posteriorly located duodenal ulcers, within 2 cm from the pylorus and typically erodes the gastroduodenal artery.

Clinical Assessment

- Presentation is by hematemesis, melena or rarely, in severe bleeding, bleeding per rectum.
- Hypovolemic shock, in severe cases.
- Abdominal examination is normal

Management

- Assessment and resuscitation
 1. Shocked patients should receive prompt volume replacement and anti-shock measures.
 2. Red blood cell (RBC) transfusion should be considered after loss of 30% of circulating blood volume
 3. Correction of coagulopathy
- Insertion of a nasogastric (NG) tube and lavage.
- Although spontaneous cessation of bleeding occurs in 70% of patients, *endoscopy* is still indicated as the first-line investigation in order to identify and often treat the bleeding point. It should be performed within 24 hours of the initial presentation, especially in patients with hypotension, age older than 60 years and multiple co-morbidities.
- Laboratory findings: anemia
- Imaging: non-specific

Major Endoscopic Stigmata of Recent Hemorrhage (Table 1)

Table 1: Endoscopic stigmata of recent hemorrhage

Stigmata	Risk of Rebleeding
1. Pulsatile or oozing hemorrhage	>85%
2. Fresh clot on ulcer	85%
3. Adherent clot on ulcer	50%
4. Visible vessel in base of ulcer	40%

Endoscopic Control of Hemorrhage

In the presence of major signs of recent hemorrhage, endoscopic therapy significantly reduces the risk of re-bleeding. Methods include the following:

- Injections
 1. Adrenaline: typically, a 0.5-1 ml of diluted adrenaline (1:100000) into each of the four quadrants of the ulcer base.
 2. Ethanol
 3. Variceal sclerosants (sclerosing agents)
- Heater probe
- Clips
- Bipolar electro-coagulation
- Nd-YAG laser

Pharmacological Therapy

In conjunction with endoscopic therapy, pharmacological therapy is started:

- Acid suppression by proton-pump inhibitors (PPIs) to maintain intra-gastric pH > 6 to stabilize clots and prevent rebleeding. Omeprazole 80 mg bolus injection is administered followed by 8 mg/hour intravenous (IV) infusion for 72 hours.
- Tranexamic acid
- Eradication H-pylori infection

Indications of Emergency Surgical Treatment of a Bleeding PU

- Continuous bleeding e.g. spurting vessel at endoscopy that does not stop with adrenaline
- Rebleeding in hospital
- Transfusion requirements of 4-6 units in 24 hours

Principles of Emergency Surgical Treatment of a Bleeding PU

Bleeding duodenal ulcer

- Exploratory midline incision
- Duodenotomy
- Stop bleeding by under-running the bleeding vessel with a suture.
- Ligation of the gastroduodenal artery
- Definitive treatment by truncal vagotomy and pyloroplasty if the patient's condition permits.

Bleeding gastric ulcer

- Antrectomy and gastro-jejunostomy to prevent re-bleeding and exclude malignancy.

PERFORATION

Perforation of PU is usually related to non-steroidal anti-inflammatory drug (NSAID) intake. It usually occurs in the anterior aspect of duodenal ulcers. It is more common in males than females, and more common in duodenal ulcers than gastric ulcers.

Clinical Features (Peritonitis and Sepsis)

- Sudden severe epigastric pain (acute abdomen) then becomes diffuse abdominal pain.
- Starts as chemical peritonitis. After 12 hours, bacterial peritonitis develops.
- Sepsis: fever, tachycardia
- Abdominal examination: tenderness, rebound tenderness, board-like rigidity, silent abdomen.
- 50% of patients have no history of peptic ulcer disease.
- The perforation may become sealed by omentum and intestinal loops and the condition improves. These patients can be managed conservatively.
- Fluid may track along the right paracolic gutter to the right iliac fossa causing pain in the right iliac fossa

Investigations

- Laboratory findings: leukocytosis
- Plain X-ray abdomen standing: free air under right copula of diaphragm (Figure 1).
- Ultrasound (US): large amount of free intra-peritoneal fluid collection.
- Computed tomography (CT) scan: pneumo-peritoneum.
- Endoscopy and barium meal are *contraindicated*.

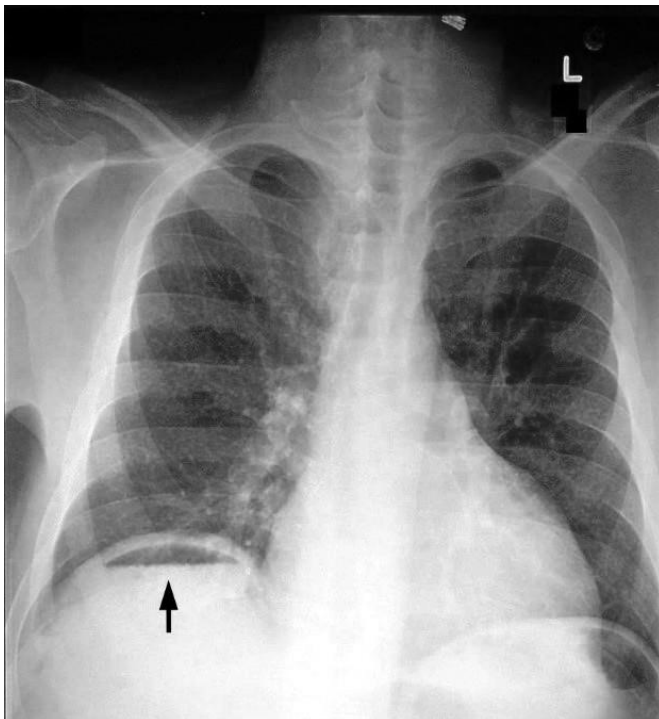


Figure 1: Plain X-ray abdomen, erect position, showing air under right copula of diaphragm (arrow)

Pre-operative Preparation

- Nothing per mouth
- Vigorous fluid resuscitation
- Monitoring urine output
- Correction of electrolyte imbalance
- Nasogastric (NG) tube
- Broad-spectrum Antibiotics
- Analgesia e.g. opioids
- Central venous pressure (CVP) measurement
- Proton pump inhibitors (PPIs)

Surgical Treatment of Perforated Duodenal Ulcer

- Laparotomy: Midline exploratory incision
- Peritoneal lavage/toilets to remove all toxic fluid and food debris.
- Omental (Graham) patch to reinforce the approximated edges of the ulcer (Figure 2)
- The operation can be performed laparoscopically.

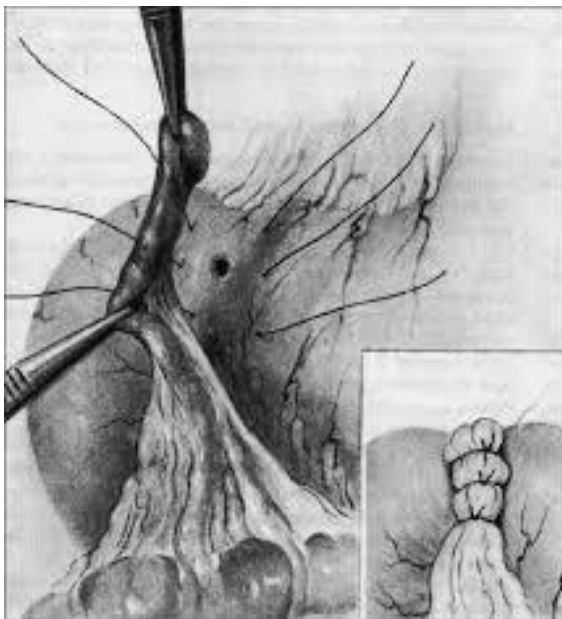


Figure 2: Omental patch for treatment of perforated duodenal ulcer

Surgical Treatment of Perforated Gastric Ulcer

Options include the following:

- Simple closure
- Wedge resection
- Biopsy
- Antrectomy and gastro-jejunostomy

GASTRIC OUTLET OBSTRUCTION (PYLORIC STENOSIS)

In a chronic process, duodenal ulcer can cause stricturing and narrowing of the first part of the duodenum, leading to gastric outlet obstruction, but an obstructing tumor is a much more common cause nowadays.

Clinical Presentation

- Long history of peptic ulcer disease
- Projectile non-bilious vomiting of undigested bad odor food.
- Epigastric pain
- Weight loss and dehydration
- Electrolyte disturbances:
 1. Hypochloremic metabolic alkalosis
 2. Hypokalemia
 3. Hypocalcemia and tetany.
- *Inspection:* Visible peristalsis
- *Palpation:*
 1. Palpable distended stomach
 2. Succussion splash

Investigations

- *Laboratory findings:* anemia, hypo-albuminemia, metabolic alkalosis, hypokalemia, elevated urea and creatinine levels.
- *Imaging*
 1. Barium enema reveals a markedly distended stomach that may reach the pelvis with a characteristic soap-dish appearance (Figure 3).
 2. CT scan with IV contrast
- *Endoscopy and biopsy:* usually the scope cannot pass to the duodenum.



Figure 3: Barium meal showing markedly dilated stomach (soap-dish appearance)

Management

- Exclusion of malignancy by CT scan, endoscopy and biopsy from the stricture.
- In mild cases, inflammation and partial obstruction may resolve with gastric lavage, acid suppression therapy and antispasmodics
- In severe cases, a hard fibrotic stricture develops causing a dilated atonic stomach.

Pre-operative Preparation

- Correction of fluid and electrolyte abnormalities by normal saline and potassium.
- Nasogastric lavage
- Proton pump inhibitors

Surgical Treatment

- Truncal vagotomy and a drainage procedure either gastro-jejunostomy (Figure 4) or pyloroplasty.
- Endoscopic balloon dilatation, but it has inferior results than surgery.

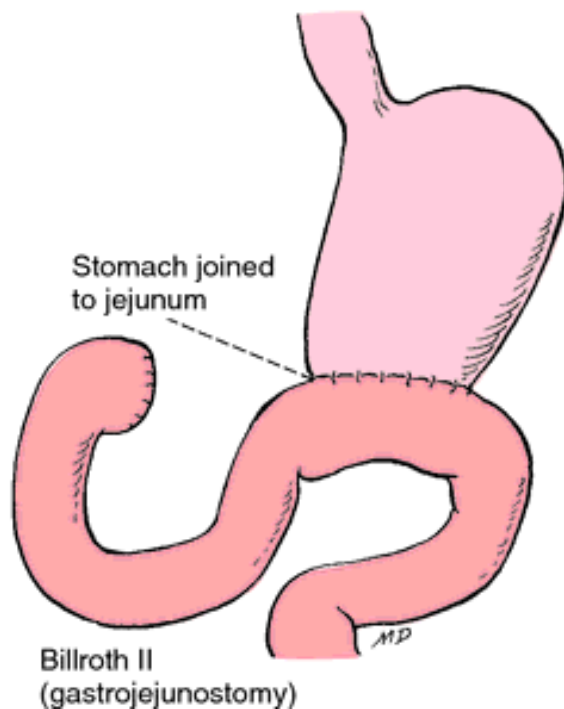


Figure 4: Simple loop gastro-jejunostomy (the stomach is joined to the jejunum)

OTHER COMPLICATIONS

1. Hour-glass deformity of the stomach
2. Recurrence of ulceration

Gastric Neoplasms

BENIGN GASTRIC TUMORS & POLYPS

The widespread use of endoscopy has increased the frequency of detection of gastric wall lesions in symptomatic & asymptomatic patients.

Benign Gastric Tumors

- 40% mucosa-based
- 40% muscularis-based

Gastric Polyps

Types

1. *Hyperplastic (regenerative) polyps*
 - These are the commonest (80%) and occur in association with gastritis and peptic ulcer (inflammatory origin).
2. *Adenomatous polyps*
 - True gastric adenomas are rare (only 5% of gastric polyps).
 - The majority lie in the antral lesion
 - May be sessile, villous, pedunculated, or lobulated.
 - May be single or multiple
 - The progression of a gastric polyp into carcinoma is very rare (contrary to what occurs in the colon).
3. *Inflammatory fibroid polyps*: rare found in the gastric antrum.
3. *Myoepithelial hamartomas* (Polyps of Peutz-Jeghers syndrome)
4. *Heterotopic pancreatic tissue*.

Treatment

- *Endoscopic excision* of gastric polyps provides a minimally invasive approach to diagnosis and treatment. Polyps < 2 cm are easily snared.
- *Larger polyps or sessile polyps* are best removed *operatively* to obtain a clear margin & complete removal
- *Malignant polyps* ⇨ treat as carcinoma

GASTRIC CANCER

Types of Gastric Cancer include adenocarcinoma (94%), lymphoma (4%) & GIST (Gastrointestinal Stromal Tumors) (1%). The typical patient with gastric cancer is a male (male-to-female ratio, 2:1) & 40-70 y (The disease is rare before the age of 40 y).

Adenocarcinoma of the Stomach

Etiology (Risk Factors)

1. *Diet (major role)*: Geographic areas with high consumption of salty & smoked foods (containing polycyclic hydrocarbons, nitrosamines). *Hypothesis*: ↑ dietary nitrates ⇒ bacterial proliferation in an achlorhydria stomach ⇒ ↑ N-nitrosamine production ⇒ stomach cancer. NB. Raw vegetables, vitamin C & antioxidants are protective
2. *Cigarette smoking*: It lowers vitamin levels.
3. *Male gender*, black race, low socioeconomic class (this applies to USA population)
4. *Occupational hazards*: miners, metal workers exposed to dusts of asbestos and wood.
5. *Blood group A* (relative risk = 1.2 times).
6. *H-pylori infection* (gram -ve microaerophilic bacterium in gastric pits) ⇒ ↑ incidence.
7. *Gastric polyps*: Villous adenomatous polyp (2%) has a malignant potential.
8. *Pernicious anemia* (10% risk of gastric carcinoma).
9. *Chronic GU*: It carries a small risk
10. *Gastrectomy* (gastric dumping syndrome) ⇒ 0-5 times the risk (15 or more years).
11. *Chronic atrophic gastritis*: It is the most researched risk factor in gastric carcinoma due to:
 - The incidence of atrophic gastritis in patients with pernicious anemia is 3-6 folds ↑.
 - Several risk factors mentioned above are thought to act by causing damage to the gastric mucosa with subsequent development of atrophic gastritis.
 - There is now increasing evidence for the progression of gastric atrophy to intestinal metaplasia, dysplasia & carcinoma in-situ.

Site of Stomach Cancer

- *Antrum/lower 3rd*: most predominant (40%)
- In the last decades, *proximal cancers* (cardia a GE junction) have risen from 10% to 30% of gastric cancers. This is because distal tumors are affected by altering the risk factors. There is also ↑ in tumors in the cardia region associated with gastro-esophageal reflux.
- *Multicentric* in 20% of cases

Spread of Gastric Carcinoma

Direct extension

- To the *omentum*, transverse colon and mesocolon and left lobe of liver.
- Extension of *antral tumors* to the duodenum.
- Extension of *upper third tumors* to the esophagus.

Lymphatic spread

- To LNs along the greater & lesser curvature, then to the celiac LNs, to the splenic hilum & root of mesentery, to retro-peritoneal & hepato-duodenal LNs & then to para-aortic nodes.

Blood spread: Mainly to the liver and peritoneal cavity.

Pathology

Gross appearance

1. Polypoid
2. Fungating
3. Ulcerative
4. Scirrhous

Histologic types (Lauren, Finnish or DIO classification)

Diffuse (D)	Intestinal (I)	Mixed (O)
Young age	Old age	have a mixed morphology between both (D & I)
No history of chronic gastritis	History of chronic gastritis	
Arise from normal gastric mucosa & not associated with intestinal metaplasia or other precancerous conditions	Surrounded with intestinal metaplasia	
No gland formation	Gastric cancer cells resemble intestinal cells with gland formation	
Infiltrates the gastric wall without forming discrete masses.	Form localized expanding or ulcerating lesions	
Less differentiated (signet ring cells)	Tendency to be well differentiated.	
Spreads by lymphatics & transmural	Spreads hematogenously	
Genetic predisposition	No genetic predisposition	
Linitis plastic, signet ring carcinoma	Fungating or polypoid masses	
Association with blood group A	No association	
Penetrates the gastric wall at an early stage	Tends to be localized with less tendency to disseminate	
Worse prognosis	Better prognosis	

Clinical Presentation

1. *Vague postprandial fullness*, indigestion & malaise.
2. *Recent dyspepsia* in a patient >50 years and receiving treatment as benign ulcer for 6-12 months without investigations.
3. *Anorexia and loss of weight* denote advanced disease.
4. *Epigastric mass* (25% of cases): It usually denotes incurability but *not* irresistibility.
5. *Anemia* (Fe deficiency type due to chronic blood loss)
6. *Complications*
 - a. *Vomiting* (gastric outlet obstruction by a pyloric tumor)
 - b. *Dysphagia* (due to a tumor in the cardia)
 - c. *Acute presentation*: hematemesis and melena (in advanced cases)
 - d. *Presentation with metastases*:
 - To the neck (*Virchow node*).
 - To the pelvis (*Blumer shelf*).
 - To the ovaries (*Krukenberg tumors*).
 - To the peritoneum (ascites).

Investigations

Laboratory

- Anemia (in 40% of cases).
- CEA (↑ in 65% of cases).
- Liver function tests (LFTs).
- Electrolyte panel.

Imaging studies

- *Barium meal*: false negative (20%) especially in ulcerating tumors.
- *Chest X-ray*: for detection of extra-gastric lesions.
- *Liver US & abdominal CT scan*: for determining the extent of the tumor, para-aortic LNs involvement & hepatic secondaries.
- *EUS*: (endoscopic US) to assess depth of the tumor (T) & detect enlarged perigastric & celiac lymph nodes (LNs)

Gastroscopy and Biopsy

- Multiple mucosal biopsies; 6 in number, should be taken

Laparoscopy

- To assess the stage and the curability of the disease.

AJCC Classification "Staging of Gastric Carcinoma"

Tumor (T)		Nodes (N)		Metastases (M)	
T1	Limited to mucosa and submucosa.	No	No metastases to LNs.	M0	No metastases.
T2	Extends to the serosa.	N1	Metastases to peri-gastric LNs within 3 cm of the 1 st tumor.	M1	Distant metastases
T3	Penetrates the serosa.	N2	To LNs > 3cms from the 1 st tumor (left. gastric, splenic, celiac LNs).		
T4	Invades the adjacent organs.	N3	LNs in porta hepatic, behind pancreatic head, root of mesentery and around middle colic vessels.		

Treatment

Surgical treatment: is the only curable treatment

Objective

Removal of the tumor + uninvolved stomach margin (minimum 6 cm) and duodenum + regional LNs (corresponding to the location of the tumor) + adjacent organs (if involved).

- **Antral tumor**: Distal gastrectomy+ omentectomy + 3-4 cm duodenum + subpyloric LNs + reconstruction with GJ or gastro-duodenostomy.
- **Body tumor and extensive tumors**: Total gastrectomy + Splenectomy (debatable) + reconstruction (pouch or Roux en Y esophago-jejunostomy).
- NB: Routine splenectomy does *not* improve survival. Thus, splenectomy *should not be performed unless* there are adhesions with the tumor to or infiltration of the spleen.
- **Cardia tumor**: Esophago-gastrectomy (via 2 incisions)

Japanese classification of gastric resections for cancer based on radicality (R)

R0	Gastrectomy with incomplete removal of N1 nodes.
R1	Gastrectomy with complete removal of N1 nodes (along the lesser & greater curvatures).
R2	Gastrectomy with removal of N2 nodes (LNs along the main arteries: left gastric, common hepatic, splenic, celiac).
R3	Gastrectomy with removal of N3 (LNs in the porta hepatic, behind the head of pancreas, root of mesentery, around the middle colic vs.).
R4	Gastrectomy with removal of N4 nodes

Palliative resection

- If life expectancy is > 2 ms, palliative gastrectomy is better to be done than palliative GJ.
- For inoperable cases of cardiac tumors: Intubation with a Celestin tube is done to overcome the dysphagia.

Adjuvant chemotherapy

- Not of much value (response is only 20%)
- Regimens used: 5 FU or Adriamycin or Mitomycin or a combination (FAM)

Non-epithelial Gastric Tumors

Gastric Lymphomas

All are non-Hodgkin type (B cell mucosa associated with MALT lymphoma)

Clinical presentation

- Bulky tumor: palpable mass (50%) when discovered versus 25% in adenocarcinoma).
- Symptoms are mild in relation to tumor size, but may present with *bleeding & perforation*.

Diagnosis

- Barium meal + CT scan (for staging)
- Gastroscopy with biopsy & brush cytology.

Treatment

- Low-grade lymphoma ⇒ Cyclophosphamide.
- High-grade lymphoma ⇒ surgical resection (this gives the best results + total abdominal radiation + *intra-operative staging* (Needle biopsy both hepatic lobes. Celiac & para-aortic LN biopsy, splenectomy if directly invaded) + post-operative chemotherapy (patients with high risks of recurrence). The 5-y survival is 50%.

GIST (Gastrointestinal Stromal Tumors)

- **Definition:** These are *mesenchymal neoplasms* derived from *interstitial cells of Cajal* (pacemaker cells controlling peristalsis).
- **Pathology:** GISTs are large tumors with high mitotic rate and tend to behave malignantly. Stomach is the commonest site.
- **Treatment:** GIST is the only type that has a real systemic medical Rx because of the expression of KIT (tyrosine kinase receptor). Clinical trials have shown partial response rates in patients treated with Imatinib for advanced, recurrent or metastatic GISTs. It has been used in the pre-operative Rx to ↓ tumor size before surgical resection

Gastric Carcinoids Extremely RARE

Small Intestinal Tumors

The small bowel is the least common site for primary cancer. The small bowel contains > 90% of the mucosal surface area of the gastro-intestinal (GI) tract, but only constitute 1.1-2.4% of all GI primary malignancies. The incidence of small intestinal tumors decreases from duodenum to ileum.

RISK FACTORS OF DEVELOPING SMALL INTESTINAL CANCER

1. Ingestion of smoked or cured foods
2. Consumption of red meat
3. Celiac sprue
4. Crohn's disease
5. Peutz-Jeghers syndrome
6. Hereditary non-polyposis colo-rectal cancer
7. Familial adenomatous polyposis (FAP)
8. Acquired immunodeficiency syndrome (AIDS)

CLASSIFICATION OF TUMORS OF THE SMALL INTESTINE

A. BENIGN TUMORS OF THE SMALL INTESTINE

Benign neoplasms of the small bowel account for 30-50% of small bowel tumors. They are most frequently encountered in the *duodenum* as incidental findings during endoscopic examinations.

Adenoma

Adenomas are the most common neoplasms of the small intestine, and are mostly encountered in the duodenum. Adenomas have different types that include:

1. *Tubular adenoma*: low malignant potential, pedunculated and presents with bleeding. Endoscopic polypectomy is the treatment of choice.
2. *Villous adenoma*: sessile lesion with high malignant potential, mostly encountered in the peri-ampullary area. Treatment is by endoscopic polypectomy for benign lesions, or surgical local excision.
3. *Tube-villous adenoma*: similar to villous adenoma
4. *Brunner's gland hyperplasia*: hyperplasia of the gland of the first part of the duodenum is a benign condition, treated by endoscopic polypectomy

Hamartoma

- Small intestinal hamartomas are usually a part of **Peutz-Jeghers syndrome**, characterized by familial intestinal hamartomatous polyposis (multiple polypi in the jejunum and ileum 25% gastric, 50% colonic), melanosis of oral mucous membrane, lips as well as palmar and plantar surfaces, intermittent intussusception, and bleeding.
- Rare malignant potential
- Increased incidence of ovarian, breast, endometrial, and pancreatic tumors.
- Treated by surgical resection or endoscopic polypectomy

Hemangioma

- Present by bleeding, occult or overt.
- Diagnosed and treated by selective angiography and surgical resection

B. PRIMARY MALIGNANT TUMORS OF THE SMALL INTESTINE

Primary malignant small intestinal neoplasms are summarized in Table 1, which also shows their incidence and predominant site.

Table 1: Primary malignant small intestinal neoplasms

Tumor	Incidence	Predominant Site
1. Adeno-carcinoma	40-50%	Duodenum
2. Carcinoid tumor	20-40%	Ileum
3. Lymphoma	15-20%	Ileum
4. Gastro-intestinal stromal tumor (GIST)	up to 15%	No regional variation

C. PRIMARY MALIGNANT TUMORS OF THE SMALL INTESTINE

- The small bowel is frequently affected by metastases either from direct extension (local invasion by cancers), intra-peritoneal carcinomatosis, or hematogenous spread.
- Common primary malignancies that can give small intestinal metastasis include malignant melanoma (commonest), ovarian, bladder, breast or lung cancer

CLINICAL PRESENTATION OF SMALL INTESTINAL TUMORS

- The prevalence of small bowel tumors identified at autopsy is 0.2-0.3%, which is significantly higher than the rate of operations for small bowel tumors.
- This suggests that the majority of small bowel tumors are *asymptomatic*.
- The most common signs and symptoms of small intestinal neoplasms include the following:
 1. Abdominal pain (67%)
 2. GIT bleeding (occult, melena or manifest bleeding) (41%).
 3. Weight loss (38%).

4. Nausea and vomiting (33%).
5. Bowel obstruction (13%).
6. Intussusception (in adults, usually benign)
7. Palpable mass (7-18%) (GIST)
8. Perforation (25%) (lymphomas)
9. Painless jaundice (peri-ampullary carcinoma)

DIAGNOSIS OF SMALL INTESTINAL TUMORS

Laboratory Tests

- Laboratory tests are non-specific, with the exception of elevated serum 5-hydroxyindole acetic acid levels in patients with the carcinoid syndrome.
- Elevated CEA levels are associated with small intestinal adeno-carcinomas, but only in the presence of liver metastases.

Imaging Studies

Plain Abdominal Radiographs

- Plain X-rays of the abdomen may show an associated small bowel obstruction, or rarely a calcified abdominal mass

Contrast (Barium) Radiography of the Small Intestine

- Contrast radiography of the small intestine may demonstrate benign and malignant lesions.
- *Enteroclysis* is reported to have a sensitivity of over 90% in the detection of small bowel tumors and is the test of choice, particularly for tumors located in the distal small bowel.
- *Upper GI with small bowel follow-through examinations* have reported sensitivity of only 30%-44%.

Computed Tomography (CT) Scan - Magnetic Resonance Imaging (MRI)

- CT scan has a low sensitivity for detecting mucosal or intra-mural lesions, but can demonstrate large tumors and is useful in the staging of intestinal malignancies. Sensitivity depends on tumor location 100% in the duodenum, 17% in the jejunum and 70% in the ileum.
- *CT enteroclysis* (helical CT with enteroclysis)

Bleeding scan and angiography

- Tumors associated with significant bleeding can be localized with angiography or radioisotope-tagged RBC scans.

Endoscopic Studies - Biopsy

- *Upper GI endoscopy*: Tumors located in the *duodenum* can be visualized and biopsied on upper GI endoscopy.

- *Endoscopic ultrasonography*: It can offer additional information such as the layers of the intestinal wall involved by the lesion.
- *Colonoscopy*: It can occasionally visualize the distal ileum successfully.
- *Intra-operative enteroscopy*: It can be used to directly visualize small intestinal tumors beyond the reach of standard endoscopic techniques.
- *Capsule endoscopy and double-balloon endoscopy* have been recently used to evaluate the small bowel.

MALIGNANT SMALL BOWEL TUMORS

Adenocarcinoma

Incidence

- *Location*: 40% in the duodenum – Peri-ampullary lesions.
- *Age*: Mean age is 60 years
- *Gender*: Male predominance (2:1)

Conditions which Increase the Relative Risk of Adenocarcinoma

1. Familial Adenomatous Polyposis (FAP) Coli and Gardner's syndrome.
2. Villous adenomas.
3. Hereditary non-polyposis colorectal cancer (Lynch II syndrome).
4. Inflammatory bowel disease (BD) (Crohn's disease and ulcerative colitis).
5. Peutz-Jeghers syndrome.

FAP and adenocarcinoma

- Patients with FAP have a nearly 100% cumulative life-time risk of developing duodenal adenomas that have the potential to undergo malignant transformation.
- The risk of duodenal cancer in these patients is over 100-fold that in the general population. Duodenal cancer is the leading cause of cancer-related death among patients with FAP who have undergone colectomy.

Crohn's disease and adenocarcinoma

- Patients with Crohn's disease have a 12 fold increased risk of small bowel cancer than normal population.
- Adenocarcinomas developing in Crohn's disease are most found in the ileum, and symptoms often mimic those of Crohn's disease.
- Risk factors of developing adenocarcinoma in a patient with Crohn's disease include (1) long duration of disease, (2) male gender, (3) fistulas, (4) surgically excluded loops of small bowel, (5) strictures, and (6) immuno-suppressive drugs.

Management - How are Adenocarcinomas Treated?

- *Resection-anastomosis*: For adenocarcinomas of the *small bowel*, a wide resection of the segment harbouring the tumor (5-cm safety margin distally and proximally) with wide excision of corresponding mesentery is done to achieve regional lymphadenectomy. Continuity of the small bowel is then restored by anastomosis.
- *Right hemicolectomy*: This is done exclusively for adenocarcinomas of the *distal ileum*.

- *Pancreatico-duodenectomy*: It is performed in case of *duodenal* tumors.
- *Palliative Surgery*: In the presence of locally advanced or metastatic disease, palliative intestinal resection or bypass (gastro-jejunostomy and choledocho-jejunostomy) is done.
- *Endoscopic stents*: Endoscopic wall stent placement for duodenal obstruction and bile stent placement for obstructive jaundice
- Chemical splanchnicectomy: Alcohol neurolysis (chemical splanchnicectomy) may be used for pain control
- Chemotherapy has no proven efficacy in the adjuvant or palliative treatment of small intestinal adenocarcinomas.

Prognosis

- Localised disease: 80% long-term survival.
- Advanced disease: 10%-15% five-year survival

Carcinoid Tumor

- Carcinoid tumors arise from *Kulchitsky cells* in the crypts of Lieberkuhn (ECL cells) – Neuroendocrine cells - (APUD system)
- They secrete several endocrine and vasoactive peptides e.g. Serotonin, Substance P, Dopamine, Histamine, Prostaglandins (E and F).
- 85% of carcinoid tumors occur in the appendix, ileum and rectum, in order of frequency, and 30-40% have multiple synchronous lesions in the jejunum
- The primary tumor is usually mostly *asymptomatic*, discovered incidentally
- They may metastasize to the liver causing **carcinoid syndrome**; reddish blue cyanosis, flushing attacks, diarrhea, asthmatic attacks, pulmonary and tricuspid stenosis, right-sided heart failure.
- *Diagnosis* is achieved by the demonstration of increased level of 5-HIAA in urine.
- Non-carcinoid synchronous tumors are found in up to 25% of patients (typically tumors of the breast, lung, stomach, or colon).
- Approximately, 25-50% of patients with carcinoid tumor-derived liver metastases will develop manifestations of the carcinoid syndrome
- *Treatment*
 1. For *small bowel*: surgical resection
 2. For the appendix: appendicectomy (appendectomy)
 3. For metastatic disease: enucleation or liver resection
 4. Somatostatin analogue: Octreotide.

Small Intestinal GIST

- The terms leiomyoma and leiomyo-sarcoma have been replaced by a more general term, gastro-intestinal stromal tumor (GIST).
- GISTs arise from the interstitial cell of *Cajal*, an intestinal pace-maker cell (theory).
- GISTs may have myogenic and/or neural features, or they may lack differentiation.
- They can present with benign or malignant characteristics.
- *Tumor markers* for GIST are CD 117 and CD 34

Management of Small Intestinal GIST

- Surgical resection of the entire tumor, along with local extensions and localized peritoneal or liver metastases, is the treatment of choice.
- If diagnosis is known pre-operatively, segmental intestinal resection can be performed and wide lymphadenectomy can be avoided as GISTs are rarely associated with lymph node metastasis
- GISTs are resistant to radiotherapy and chemotherapy, their role in adjuvant treatment is not defined.
- The specific treatment of GIST is Tyrosine Kinase inhibitor (Gleevec) Imatinib, which is used with those with advanced or metastatic disease.

Important Markers of Prognosis

- Tumor size.
- Mitotic activity
- Spread of the disease outside the bowel wall.
- Poor histological grade.
- Short time from symptoms to diagnosis.

Small Intestinal Lymphoma

- Lymphoma may involve the small intestine as primary neoplasms without systemic nodal disease or as secondary tumors as a part of disseminated lymphoma.
- Primary GI lymphomas are the most common of extra-nodal lymphomas; most commonly located in the ileum (contains the highest concentration of lymphoid tissue).
- Although partial small bowel obstruction is the most common mode of presentation, 10% of patients with small intestinal lymphoma present with bowel perforation.
- Most are Non-Hodgkin's B-cell lymphomas with intermediate or high grade features.

What defines primary lymphoma of the gut ?

1. Absence of peripheral lymphadenopathy
2. Absence of mediastinal lymphadenopathy
3. Normal white blood cell (WBC) count
4. Absence of bone marrow lymphoma
5. At laparotomy, the disease is restricted to the bowel and proximal lymph nodes with no involvement of the liver or the spleen

Conditions with impaired mucosal barrier increase the risk of small bowel lymphoma?

1. Mal-absorptive or inflammatory conditions e.g. celiac sprue, Crohn's disease.
2. Helicobacter pylori infection → small bowel lymphoma
3. Immunosuppressed patients due to:
 - Pharmacological e.g. post-transplant immunosuppressants e.g. cyclosporine
 - HIV infection (AIDS): GI lymphomas (B-cell non-Hodgkin's lymphoma) and Kaposi's sarcoma can appear in any organ.
 - Diseases that cause immunological dysfunction e.g. rheumatoid arthritis
 - Congenital e.g. agammaglobulinemia and congenital immunodeficiency syndrome

Types of primary intestinal lymphoma

They are Non-Hodgkin's lymphoma (NHL), mainly of the B-cell type:

1. Mature (peripheral) B-cell

- Extra-nodal Marginal Zone B-cell lymphoma → (MALT Type).
- Diffuse large B-cell → High-grade and MALT type.
- Mantle cell.
- Burkitt's lymphoma

2. Mature T cell

- Enteropathy type T-cell (EATCL)

Management of Small Intestinal Lymphomas

- *Localized* small intestinal lymphoma should be treated with *segmental resection* of the involved intestine and adjacent mesentery.
- If the small intestine is *diffusely* affected by lymphoma, *chemotherapy* rather than surgical resection should be the primary therapy.
- Surgical resection for stage I or II lymphoma, debulking for large complicating tumors in stage III or IV
- All patients receive *chemotherapy*
- The role of *radiation* therapy is unclear

Bariatric Surgery

MORBID OBESITY – CONCEPTS AND DEFINITIONS

Obesity has become one of the most widespread health problems in the world. Prevalence of obesity in USA increased from 12% to 21% between 1991 and 2001 = 15 million people. Obesity is the 2nd most common cause of death from a modifiable behavioural risk factor

Who is Considered Obese ?

Body Mass Index (BMI) Concept

Weight measurement alone is *not* ideal for determination if the patient is obese or not; that is why the Body Mass Index (BMI) concept is used.

- BMI is defined by the ratio of an individual's height to weight [BMI = body weight (Kg) / height (m²)].
- *Normal*: The normal BMI ranges from 20-25.
- *Morbid obesity*: An individual is considered morbidly obese if he or she has a BMI of 40 or more, or 35 or more and experiencing obesity-related health conditions, such as diabetes mellitus (DM) and hypertension (HTN).
- *Super-obesity*: Persons with BMI above 50 are considered Super Obese. Super-obesity should be dealt with as a special entity because of the following:
 1. They suffer much more from obesity-related health problems.
 2. They impose much higher risk to surgery and anesthesia.
 3. They need special beds, chairs, trolleys, surgical tables and surgical instruments with extra-lengths.
 4. They are better to be dealt with at dedicated institutions.

Apples versus Pears Concept

Not only the absolute value of BMI is important, but also the distribution of fat. Having an apple-shaped body means that you carry a higher proportion of fat around your abdomen, whereas a pear-shape indicates you carry most of fat around your legs and hips (Figure 1).

- For example, if you have a bit of a beer belly or a spare tyre around you, you are more of an apple shape! If you are wider around the hips you are a pear-shape. In general, being more pear-shaped means you are at lower risk of chronic disease than being an apple-shape and this mainly has to do with your visceral and subcutaneous (SC) fat storage.

- Visceral fat is the fat that sits deep in our abdomen and surrounds our organs, while SC fat is stored under our skins and is the fat that you can feel in the skin folds.
- The visceral fat is the most dangerous and it increases our risks of type-2 DM, cardiovascular disease, hypertension, inflammation and other diseases.

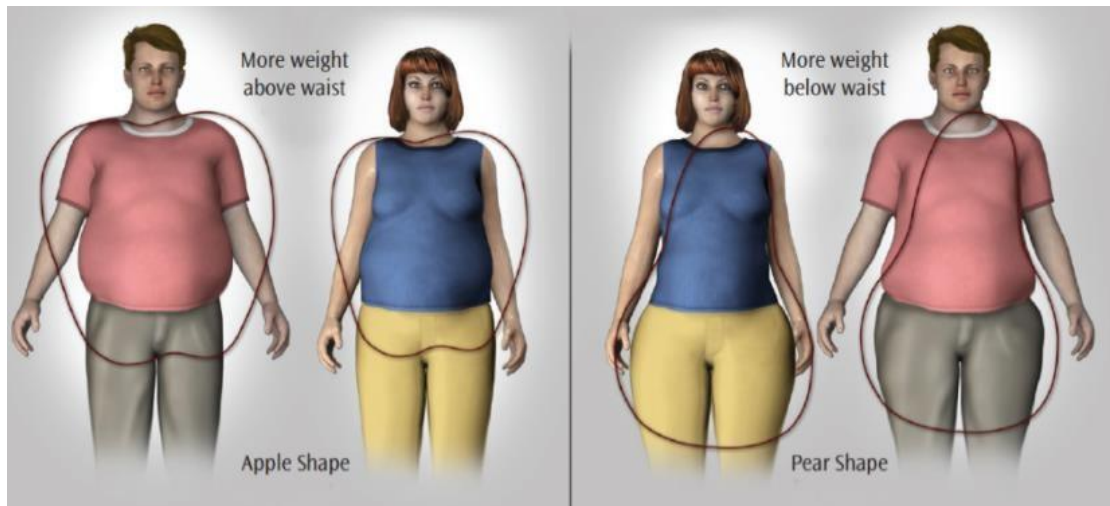


Figure 1. Apples versus pears body types

1. COMMON OBESITY-RELATED CONDITIONS

Obesity is associated with the following conditions:

1. *DM-Type 2*: People who are obese become resistant to insulin, which regulates blood glucose levels leading to Type 2 diabetes.
2. *HTN/heart disease*: The obese person usually gets HTN, which results in stroke and damage to the heart and kidneys.
3. *Osteoarthritis of weight-bearing joints*: Additional weight placed on joints, especially knees and hips can cause rapid wear and tear, along with pain and inflammation. Similarly, the strain on bones and muscles in the back leads to disk problems, pain and reduced mobility.
4. *Sleep apnea/respiratory problems*: Fat deposits in the tongue and neck can block the air passages, especially in patients who sleep on their backs. This interferes with sleep leading to daytime drowsiness and headache.
5. *Gastro-esophageal reflux disease (GERD) (hiatal hernia and heart-burn)*: Excess weight weakens and overloads the gastro-esophageal junction and thus allows stomach acid to escape into the esophagus (reflux) leading to heart-burn and acid indigestion. About 10-15% of patients with even mild heart-burn develop Barrett's esophagus (a pre-malignant change in the lining membrane and a cause of esophageal cancer).
6. *Depression*: People who are obese must deal with constant, depressing emotional challenges.
7. *Infertility*: Obesity affects male and female hormones, disrupting normal cycles and function leading to difficulty or inability to conceive.

8. *Urinary stress incontinence*: A large, heavy abdomen relaxes the pelvic muscles, compounding the effects of childbirth. This weakens the urinary bladder sphincter, allowing leakage when coughing, sneezing, or laughing.

TREATMENT OF MORBID OBESITY

There are several arms for the management of the obesity problem:

1. *Non-invasive*: Exercise and medical therapy.
2. *Invasive, non- surgical*: Intra-gastric balloon insertion
3. *Surgical*: Bariatric surgery

Non-invasive Treatment

- Diet – low in calories, fat and carbohydrates
- Exercise – 40 minutes, 5 times per week
- Behavior Modification – eat 3 sensible meals per day, avoid snacks
- Drugs/Prescription medications
 1. Appetite suppressants
 2. Anti-depressants
 3. Drugs that reduce fat absorption

Disadvantages of non-invasive therapy

1. Most patients (95-97%) regain most or all of the weight that was lost within 2-5 years following diet or drug treatment
2. The average amount of weight loss is relatively small (5-20 Kg)
3. Drug therapy may be associated with severe complications
4. Very difficult for most people to maintain these programs in the long-term
5. “Yo-Yo” effect of many different programs leads to significant weight fluctuations

Intra-gastric Balloon Insertion

- This is balloon manufactured from non-irritant material to be placed in the stomach by the aid of upper gastrointestinal (UGI) endoscopy. After its placement in the stomach, it is inflated with methylene blue solution to a volume of 500-700 cc (Figure 2).
- It decreases the functioning volume of the stomach in order to restrict food-intake.
- It aims at a decrease of 25-30 kg of the patient’s body weight
- It has the advantage of being a simple and short procedure.
- Its main disadvantage is that it must be removed after six months. Thus, most patients regain weight after its extraction.
- It is very useful as an adjuvant procedure before operating on super-obese patients in order to obtain some weight reduction, which helps in improving their.

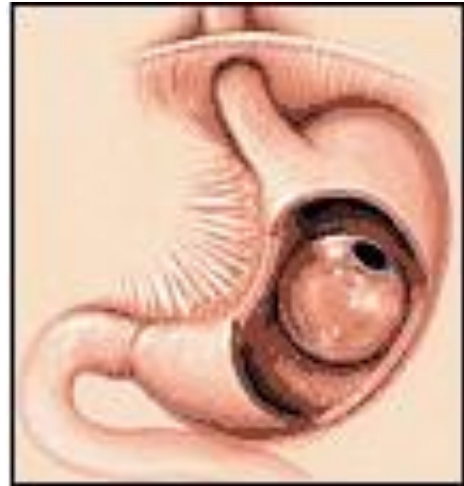


Figure 2. Intra-gastric balloon insertion for treatment of morbid obesity

Bariatric Surgical Procedures

Definition

- The name bariatric comes from the Greek words *baros* – meaning weight and *iatreia* – meaning treatment
- Bariatric surgery describes a group of operations performed primarily to reduce body weight (treat morbid obesity). This is typically achieved through inducing anatomical alteration to the gastro-intestinal tract (GIT). These operations aim at decreasing food reservoir, speeding food passage through the GIT and/or shortening the journey through the bowel.

Classification

Bariatric operations are classified according to the mechanism of action into:

1. Restrictive Operations

This group of operations works mainly by limiting food reservoir, which limits food-intake. Rapid food passage occurs in some of these operations. This, when present, acts as a 2nd mechanism.

2. Mal-absorptive Operations

Here, the main mechanism of action is to induce malabsorption through short-circuiting of the food passage via “bypassing” a variable length of the small intestine. All these operations have also a restrictive action added to it. Again, food passage is faster.

Indications

1. BMI >40
2. BMI >35 + one or more co-morbidity such as:
 - Osteoarthritis
 - Disc prolapse
 - DM type-2, dyslipidemia
 - Hypertension
 - Obstructive sleep apnea

Prerequisites for Having a Bariatric Operation

1. Age: 18-65 years (patients out of this range are only operated upon in special conditions)
2. Psychological stability and integrity.
3. Normal eating behavior
4. Normal endocrine function
5. Ability and willingness to attend for follow-up
6. Fitness for surgery and anesthesia
7. Failure of other measures of weight loss (diet and exercise)

Operative Procedures

Laparoscopic Sleeve Gastrectomy

- This is a commonly performed *restrictive* operation
- A narrow vertical pouch is designed along the lesser curvature of the stomach connecting the esophagus to the pylorus. The resulting 150 cc stomach pouch resembles a “sleeve” hence the name “sleeve gastrectomy” (Figure 3)
- *Advantages:* laparoscopic sleeve gastrectomy does *not* involve malabsorption, abnormal tracts, blind tracts, or the placement of a foreign body

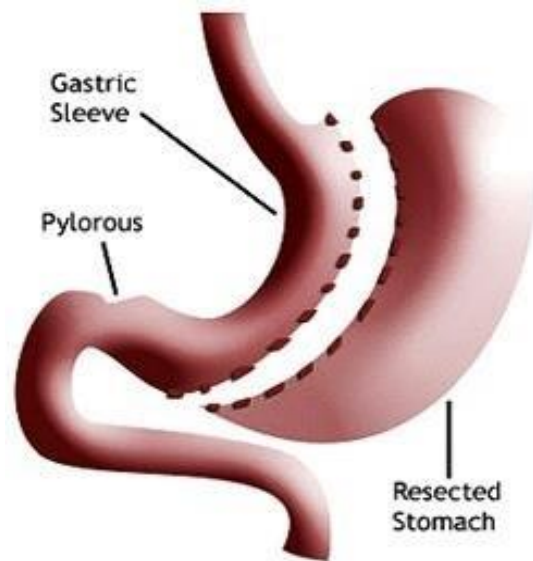


Figure 3: Sleeve gastrectomy

Laparoscopic Gastric Bypass

- These are a group of *mal-absorptive* operations in which the stomach is divided into 2 parts: a smaller proximal part that is anastomosed to the small intestine 1.5-3 meters distal to the duodeno-jejunal junction; and a larger distal part that is left attached to the duodenum through the pylorus (Figure 4)
- This “excluded” distal part of the stomach does *not* receive food. It only secretes acid and enzymes that flow distally through the duodenum to meet the food at the site of the gastro jejunostomy

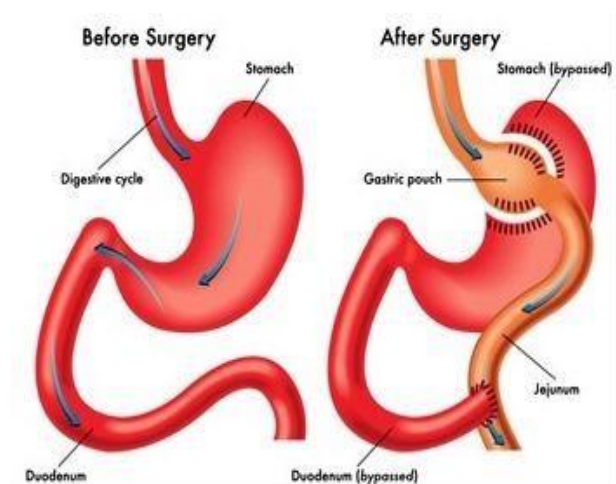


Figure 4. Laparoscopic Gastric Bypass

Post-operative Care

Post-operative instructions

- After surgery, patients should remain on a *high-protein* and *low-fat* diet and supplement their diet with *multi-vitamins*, *iron* and *calcium*, usually on a twice-daily basis.
- *Ursodiol* may be given to minimize the risk of developing gallstones during the period of acute weight loss.
- Patients must modify their *eating habits* by avoiding chewy meats and other foods that may inhibit normal emptying of their stomach pouch.

Post-operative tests

- *Nutritional and metabolic blood tests* must be performed on a frequent basis; at 6 months and 12 months after surgery, and then annually thereafter.

Post-operative body contouring

- Massive weight loss is associated with negative consequences for the body, such as flabby skin, abdominal skin overhang and pendulous breasts.
- The excess skin does *not* regain the tightness it had before the weight gain. Redundant rolls of tissue may also be associated with intertrigo and significant hygiene problems.
- Surgical correction of these body deformities can significantly enhance physical and physiological changes. Examples include abdominoplasty, buttock lift, thigh lift, mastopexy, etc.
- The usual time lapse between gastric bypass and plastic surgery procedures is 12-18 months.

Complications of Bariatric Surgery

1. Heavy patients are prone to certain hazards due to their weight during transport, positioning on the operating table and during endotracheal intubation.
2. Delayed recovery from anesthesia with possible need for post-operative mechanical ventilation.
3. Increased risk of deep vein thrombosis (DVT).
4. Post-operative hemorrhage from the stapled cut-edge of the stomach.
5. Leakage of gastric or intestinal contents through a defective staple line or sutured anastomosis.
6. Intra-operative injury to the esophagus or spleen.

Acute Abdomen and Peritonitis

DEFINITION OF ACUTE ABDOMEN

- Acute abdomen: Any serious acute intra-abdominal condition attended by sudden, spontaneous severe pain in the abdomen that is less than 24 hours in duration.
- Acute surgical abdomen: Pain arising in a previously apparently healthy person which lasts more than six hours.

ETIOLOGY OF ACUTE ABDOMEN

Surgical Causes of Acute Abdomen (Figure 1)

Gastro-intestinal Tract (GIT) Disorders

- Non-specific abdominal pain (43%)
- Appendicitis (24%)
- Small and large bowel obstruction (5%)
- Perforated peptic ulcer (3%)
- Bowel perforation
- Diverticulitis (2%)
- Gastritis and gastroenteritis

Liver, Spleen and Biliary Tract Disorders

- Acute cholecystitis (9%)
- Acute cholangitis
- Hepatic abscess
- Ruptured hepatic tumour
- Spontaneous rupture of spleen
- Splenic infarct

Pancreatic Disorders

- Acute pancreatitis (2%)

Urinary Tract Disorders

- Renal colic (4%)
- Renal infarct

Gynecological Disorders

- Ruptured ectopic pregnancy
- Twisted ovarian tumour
- Ruptured ovarian follicle cyst
- Pelvic inflammatory disease (PID)

Vascular Disorders

- Ruptured aneurysm
- Mesenteric vascular occlusion

Peritoneal Disorders

- Peritonitis
- Intra-abdominal abscess

Post-operative Causes

- Bleeding
- Anastomotic leak
- Bowel obstruction
- Abscess
- Abdominal compartment syndrome

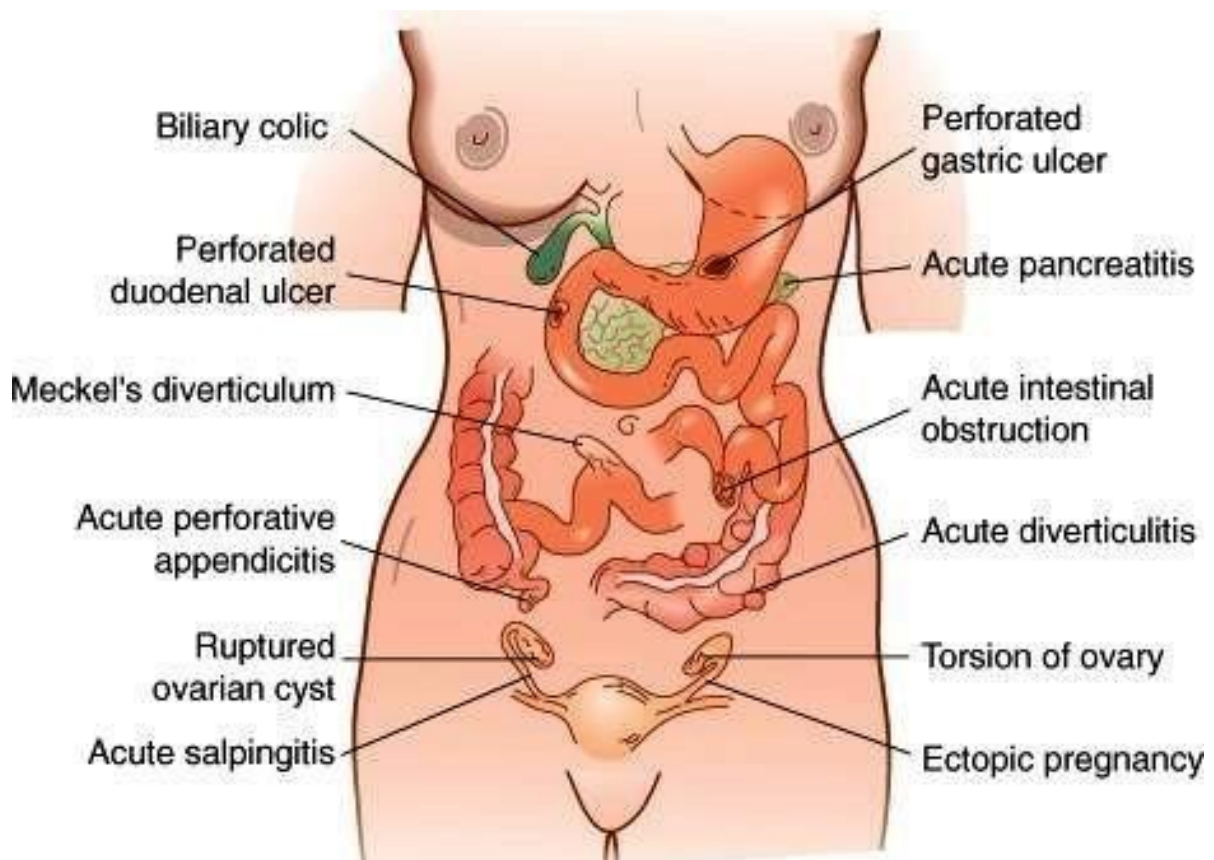


Figure 1. Common surgical causes of acute abdomen

Medical Causes of Acute Abdomen

Endocrine and Metabolic Disorders

- Uremia
- Diabetic crisis
- Addisonian crisis
- Acute intermittent porphyria
- Hereditary Mediterranean fever

Hematological Disorders

- Sickle cell crisis

Toxins and Drugs

- Lead poisoning
- Food poisoning

Infections and Inflammatory Disorders

- Henoch-Schonlein purpura
- Systemic lupus erythematosus (SLE)
- Polyarthrititis

Tropical Diseases

- Typhoid
- Malaria
- Tuberculous (TB) peritonitis
- Amebiasis
- Worm infestations

Cardiac / Pulmonary Disorders

- Pleurisy
- Pneumonia
- Cardiac disease (e.g. myocardial infarction - MI)

Nervous System Disorders

- Disease of spine affecting nerve roots

Common Acute Abdominal Conditions in Infants and Children

- Acute appendicitis
- Non-specific mesenteric adenitis
- Intussusception
- Intestinal volvulus
- Meckel's diverticulitis

CLINICAL DIAGNOSIS

History-Taking / Symptoms

Abdominal pain

- Characterizing the pain is the key: onset, duration, location, character, etc.
 1. Visceral pain → dull, poorly localized (distension, inflammation or ischemia)
 2. Parietal pain → sharper, better localized
 3. Kidney / ureter → flank pain
- Onset
 1. Sudden (hemorrhage, perforation or torsion)
 2. Gradual (inflammation)
- Progression of pain (Figure 2)

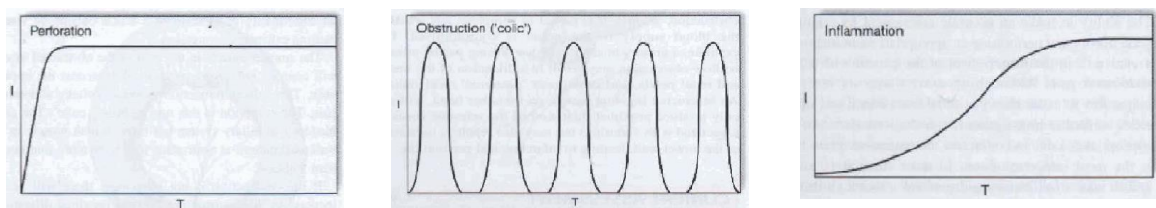


Figure 2: Progression of pain in case of perforation, obstruction (colic) and inflammation

- Referral of pain
The different sites of referred pain from various pathologic disorders are illustrated in Figure 3

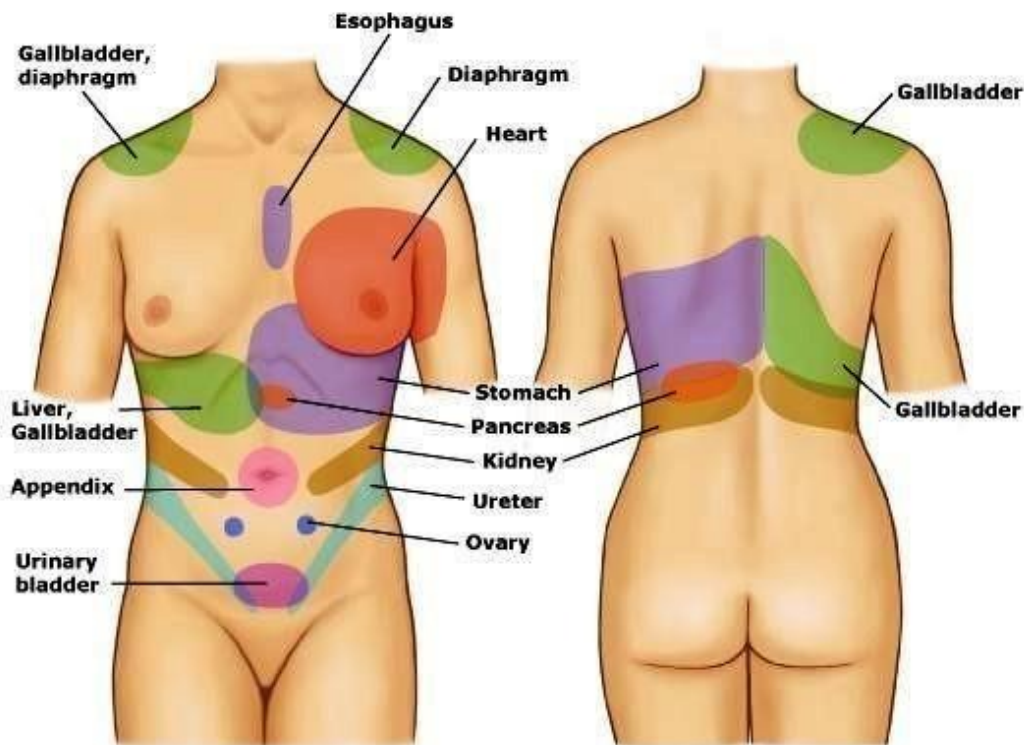


Figure 3: Sites of referred pain from different pathologic disorders

GI symptoms

1. Nausea, vomiting (bilious or bloody)
2. Constipation is the rule in most acute abdomen (due to ileus or obstruction)
3. Diarrhea (?? bloody)
4. Both, nausea and diarrhea? Medical?
5. Change in symptoms with eating?

Constitutional symptoms

1. Fever
2. Chills
3. Jaundice (passage of clay stools and dark urine denotes obstructive jaundice)

History-Taking

- Non-steroidal anti-inflammatory drug (NSAID) use (e.g. for duodenal ulcer)
- Drinking alcohol (pancreatitis)
- Prior surgeries (adhesions...)
- Hernia (obstruction, strangulation)
- Urine output (dehydration)

Common Causes of Pain According to Site

The following is an illustrated summary of the different causes of pain according to the site (location) of pain, whether generalized or localized (Figures 4-12).

- Perforation
- Mesenteric ischemia
- Abdominal Aortic Aneurysm (AAA)
- Acute pancreatitis
- DM

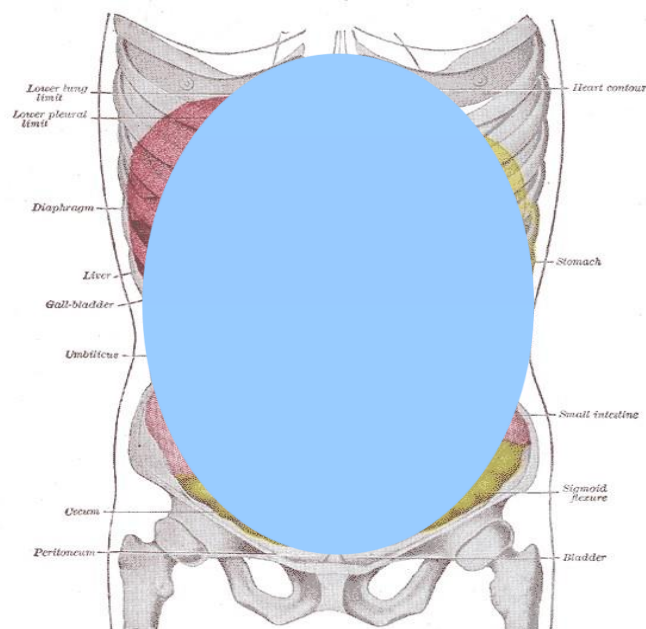


Figure 4. Common causes of diffuse (generalized) abdominal pain

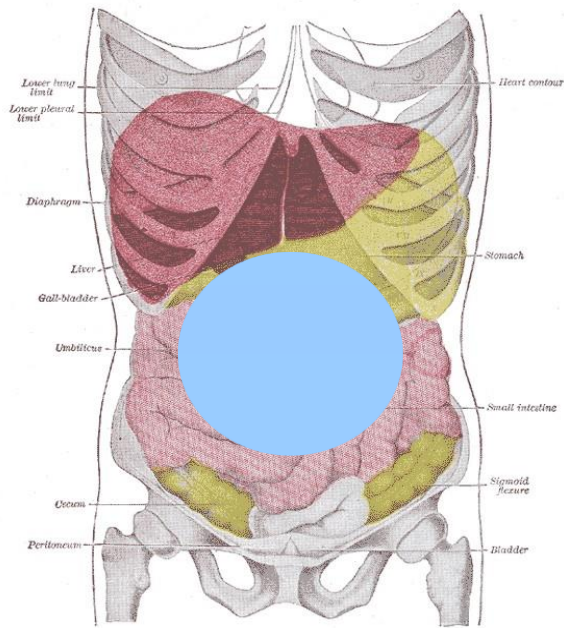


Figure 5: Causes of central abdominal pain

- Early appendicitis
- Small bowel obstruction
- Acute pancreatitis
- Ruptured AAA
- Mesenteric thrombosis
- Acute gastritis

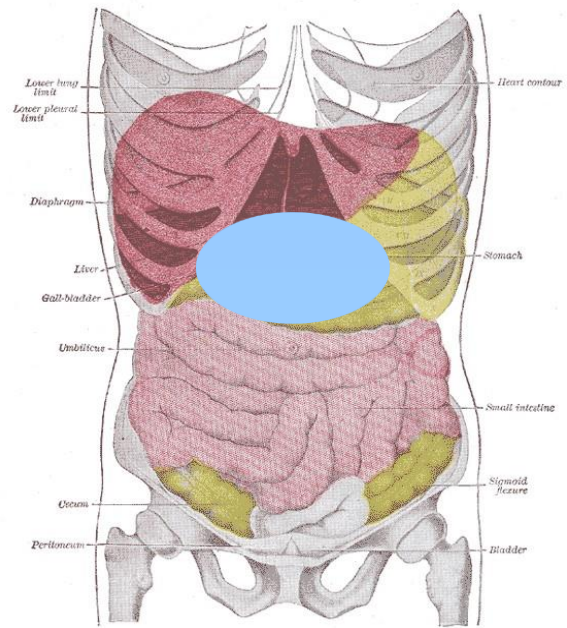


Figure 6: Causes of epigastric pain

- Duodenal ulcer (DU)
- Gastric ulcer (GU)
- Esophagitis
- Acute pancreatitis
- AAA

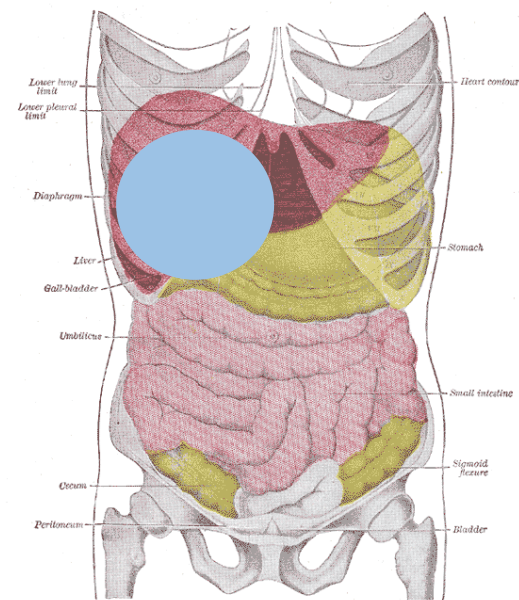


Figure 7: Causes of right upper quadrant pain

- Acute cholecystitis
- DU
- Acute pancreatitis
- Retro-cecal appendicitis
- Right pneumonia - Subphrenic abscess

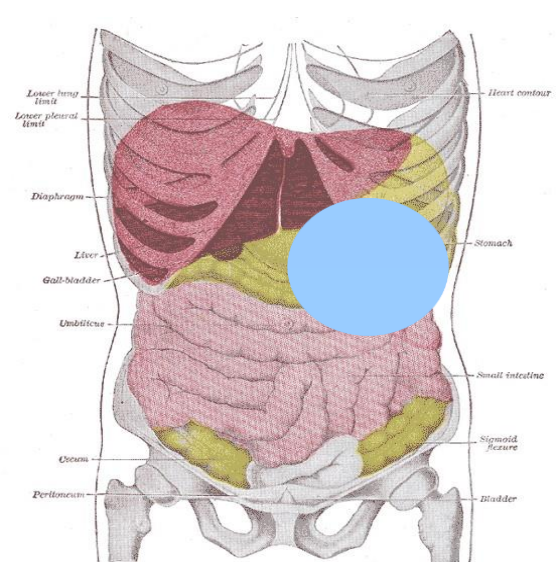


Figure 8: Causes of left upper quadrant pain

- GU
- Pneumonia
- Acute pancreatitis
- Splenic rupture - Splenic abscess
- Acute peri-nephritis - Subphrenic abscess

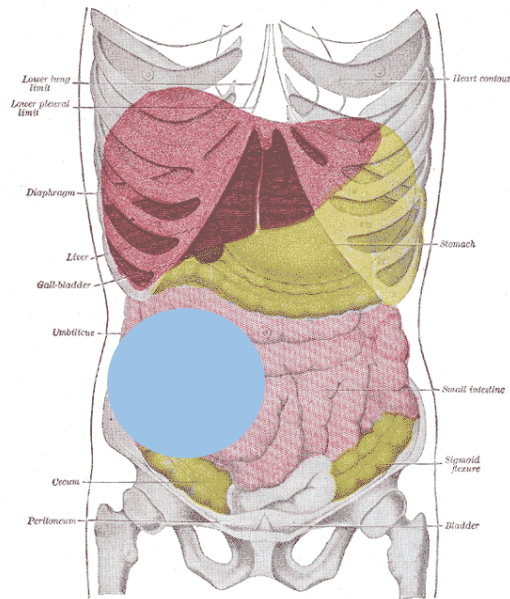


Figure 9: Causes of right iliac fossa pain

- Acute appendicitis
- Mesenteric adenitis - Meckel's diverticulum
- Perforated - DU - Diverticulitis - IBS
- PID - Ectopic pregnancy
- Ureteric colic

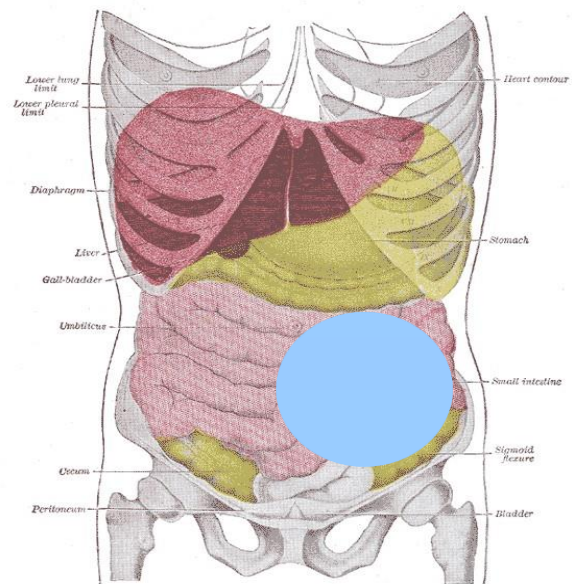


Figure 10: Causes of right iliac fossa pain

- Diverticulitis – IBS - Constipation
- PID
- Rectal Cancer
- Ureteric colic
- Ectopic pregnancy

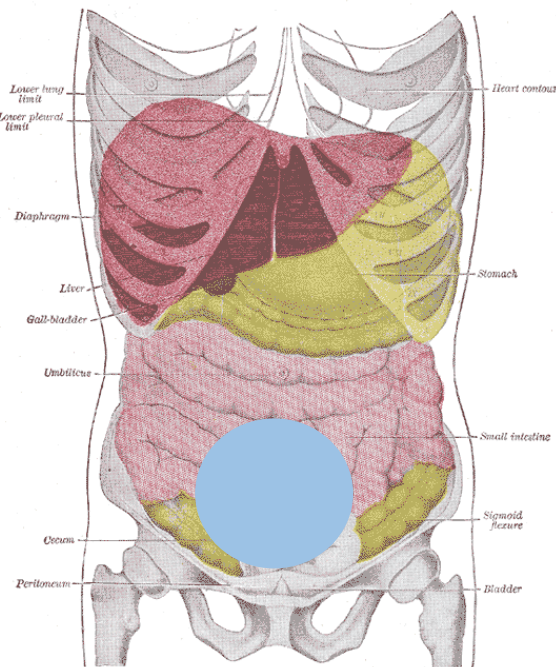


Figure 11: Causes of supra-pubic pain

- Acute urinary retention
- UTI - Cystitis
- PID
- Ectopic pregnancy
- Diverticulitis

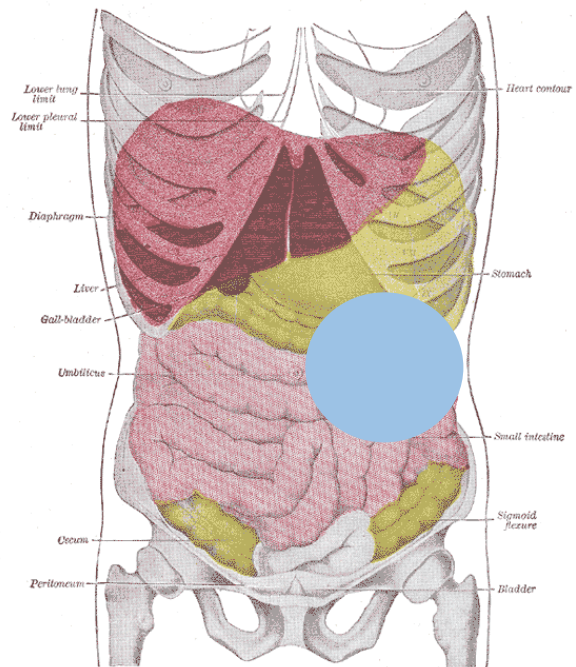


Figure 12: Causes of loin pain

- Muscle strain
- UTIs
- Renal stones
- Pyelonephritis

Clinical Examination

- General examination
- Vital signs
- Abdominal examination
- Examination of hernial orifices
- PR examination
- Scrotal examination.

INVESTIGATIONS

Laboratory Test

1. *Complete blood count (CBC)*: Leukocytosis
2. *Electrolytes*: Vomiting causes acidosis and dehydration
3. *Amylase*: Pancreatitis
4. *Liver function tests (LFTs)*: Jaundice – hepatitis
5. *Urine analysis*: UTI – stones - hematuria

Imaging Studies

1. *Plain X-ray (supine and erect)*: Obstruction – free air – stones
2. *Ultrasound*: Cholecystitis – gynecological pathology – collection
3. *CT scan*: Diagnostic
4. *Angiography*: Ischemia - bleeding

TREATMENT

Primary Care

- Monitoring vital signs and pain assessment
- Foley's urinary catheter
- Nothing orally (NPO) until surgical intervention is ruled out
- Nasogastric (NG) tube with low continuous suction
- IV fluids or nutritional support
- Analgesia for pain

Decision to Operate (Indications of Surgery)

1. Peritonitis: Tenderness, rebound tenderness, involuntary guarding
2. Severe / unrelieved pain
3. "Vitally unstable" (hemorrhage or sepsis): Tachycardia, hypotensive, white count
4. Intestinal ischemia, including strangulation
5. Pneumo-peritoneum
6. Complete or "high grade" obstruction

Individual Causes of Acute Abdomen

Acute Appendicitis (Figure 13)

- Any age
- Shifting pain
- Pain before vomiting
- Clinical picture depends on the site of appendix
- High WBC count with shift to the left
- Ultrasound to exclude other pathology especially in females
- Treatment: Appendectomy

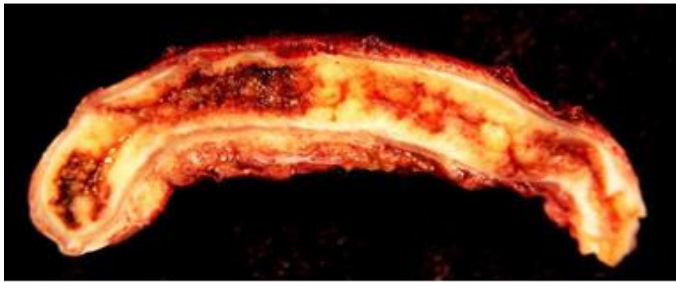


Figure 13: Acute inflammation of the appendix (acute appendicitis)

Perforated Peptic Ulcer

- Sudden onset
- Commonly in the duodenum (DU > GU with a ratio of 6:1) (Figure 14)
- Ulcer may be acute or chronic
- Board like rigidity
- Plain X-ray (KUB) shows air under diaphragm (pneumo-peritoneum) (Figure 15)
- History of a precipitating factor

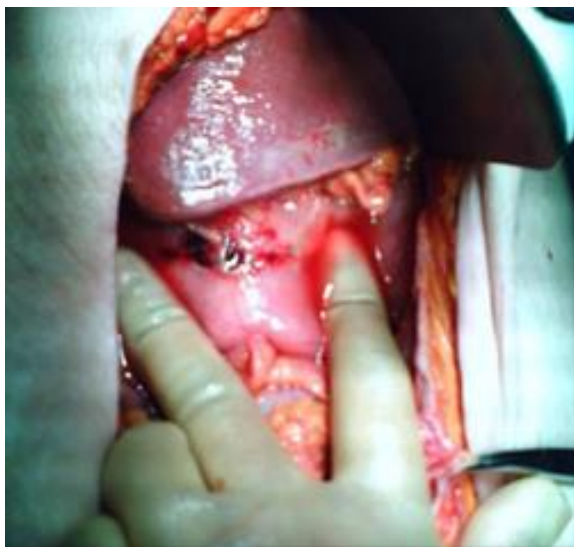


Figure 14: Perforated DU (arrow)



Figure 15: Air under diaphragm (arrow)

Acute Cholecystitis

- Diffuse or colicky right hypochondrial pain
- Overlying tenderness and rigidity
- Difficult to feel the gall bladder (GB)
- Ultrasound (US): cholecystitis – gall stones (Figure 16)

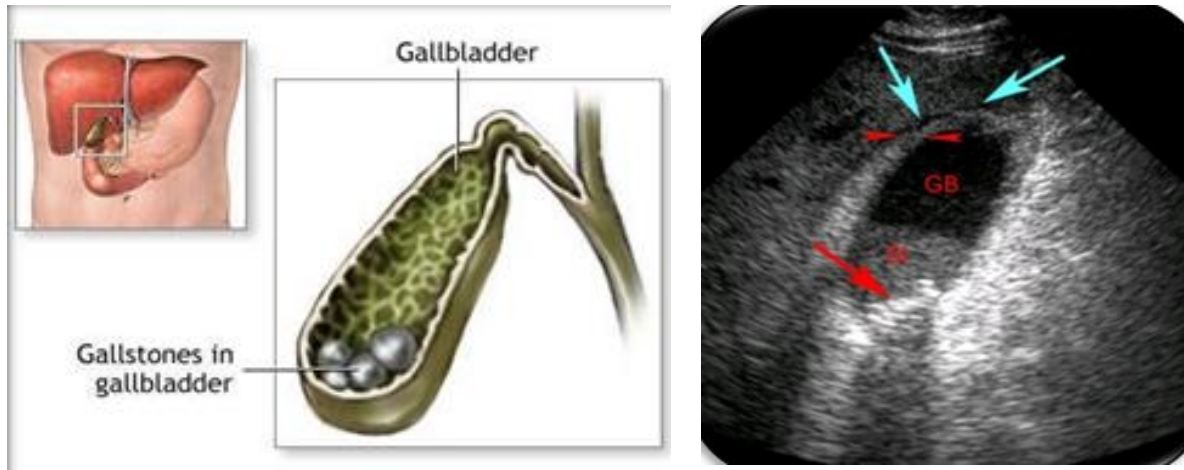


Figure 16: Cholecystitis and gallstones seen on ultrasound (arrows)

Acute Pancreatitis

- Severe epigastric pain
- Pain is referred to back
- Profuse vomiting
- The patient may be shocked
- Serum amylase within 12 hours (elevated)
- Serum lipase (elevated) is more diagnostic
- US and CT scans are diagnostic

Acute Intestinal Obstruction

- *Common causes:* Hernia - adhesions – intussusception – cancer – volvulus
- Symptoms according to site and cause of obstruction
- In general the higher up the gut, the more severe the symptoms
- Pain very severe referred to epigastrium, umbilical or hypogastrium
- Clinically- distension, features of shock
- Plain X-ray: Multiple fluid levels (step-ladder appearance) (Figure 17)

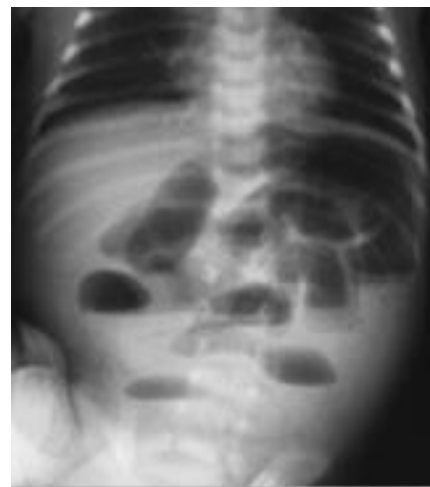


Figure 17: Plain X-ray showing multiple fluid levels

Obstruction Proximal Small Intestine

- Vomiting very early, frequent and violent, green and bilious
- Distension is *not* an early feature

Obstruction Distal Small Intestine

- Pain is less severe than proximal small bowel obstruction
- Vomiting and distension delayed

Large Bowel Obstruction

- Distension is an early feature except in intussusception
- Pain less acute, shock and vomiting late.
- Can be due to strangulation of bowel where tenderness on applying pressure is positive.
- Obstruction can be due to volvulus, colon cancer, impacted fecal matter, etc.

Simple Bowel Obstruction

- Colicky abdominal pain
- Vomiting
- Abdominal distension
- Absolute constipation
- Multiple air-fluid levels on plain X-ray

Strangulated Bowel Obstruction

- Severe pain *not* relieved by NG tube
- Localised tenderness and guarding
- Fever and tachycardia
- Leucocytosis

Acute Diverticulitis

- Rare before 40 years
- Sigmoid commonly affected (Figure 18)
- Diagnosis relies on clinical presentation
- Gastrograffin enema may be helpful
- CT scan is diagnostic (Figure 19)

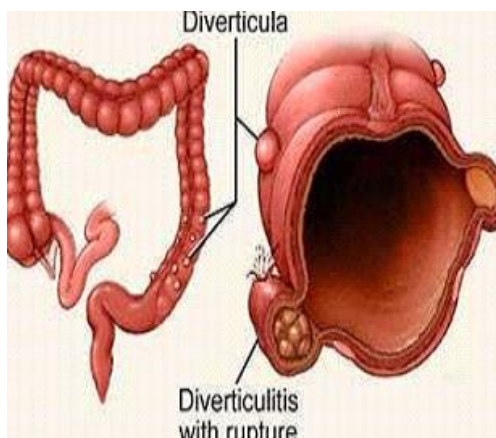


Figure 18: Diverticulitis



Figure 19: CT showing diverticulitis (arrows)

Vascular Acute Abdomen

- Acute mesenteric ischemia: Usually acute occlusion of the superior mesenteric artery (SMA) by a thrombus or an embolus (Figure 20)
- Chronic mesenteric ischemia: The patient is typically a smoker, vasculopathy with severe atherosclerotic vessel disease
- Ischemic colitis
- Any inflammation, obstructive, or ischemic process can progress to perforation
- Ruptured abdominal aortic aneurysm (AAA)

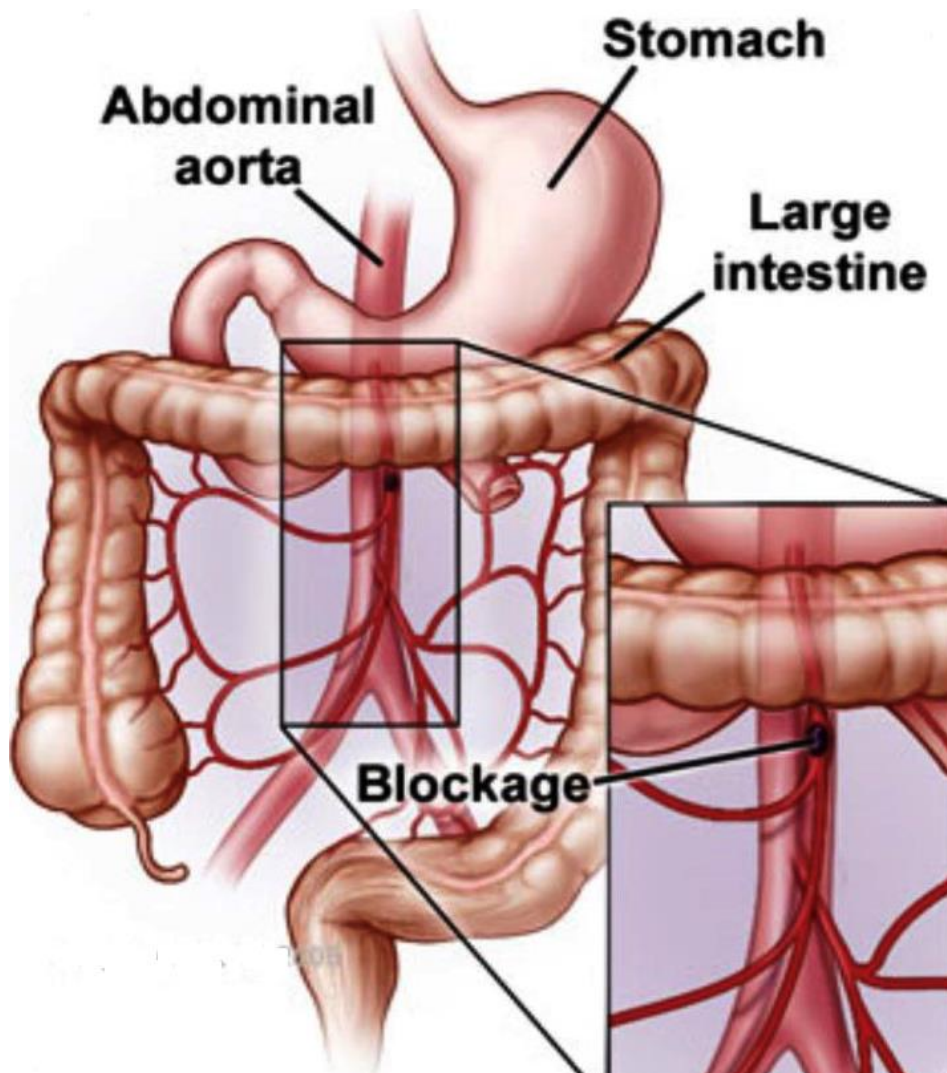


Figure 20: Acute mesenteric ischemia (inset)

Mesenteric Ischemia

- Patient is over 50 years, with valvular or atherosclerotic heart disease
- Arrhythmia or history of myocardial infarction (MI).
- Bleeding per rectum may be the early manifestation
- CT scan is diagnostic

Intestinal Obstruction

Definition

Intestinal Obstruction means that the luminal contents cannot pass through the gut tube.

CLASSIFICATION / ETIOLOGY

Intestinal obstruction has several types and can be classified in several ways:

1. Acute, sub-acute & chronic or acute-on-top of chronic:

- *Acute obstruction* = Small bowel obstruction (SBO)
- *Chronic obstruction* = Large bowel obstruction (LBO)
- *Acute-on-top of chronic* = obstruction start as large-bowel and progress as small-bowel due to yielding of ileo-cecal valve

2. Dynamic & adynamic (according to peristalsis)

- **Dynamic (mechanical) obstruction** means that luminal contents *cannot* pass through the gut because the lumen is blocked (Dynamic = audible, loud & noisy peristalsis).

Causes of dynamic intestinal obstruction:

Remember that in dynamic obstruction there is peristalsis working against an obstructing agent trying to overcome it (this peristalsis is always heard by auscultation & could be seen on inspecting the abdomen as a wave-like movement). This obstructing agent could be in the wall, in the lumen or outside the wall

- a. *Intra-luminal* (in the lumen): e.g. impacted feces, bezoars, gallstones, meconium
 - b. *Intra-mural* (in the wall): e.g. malignant (Figure 1) or benign stricture such as Crohn's disease (Figure 2), radiation stricture, post-ischemic stricture)
 - c. *Extra-mural* (outside the wall) (Figure 3): e.g. bands/adhesions, hernias (Figure 4), volvulus (Figure 5), intussusception (Figure 6), or carcinomatosis
- **Adynamic (functional) obstruction** means that luminal contents *cannot* pass through the gut tube because of disturbances in gut motility that prevent coordinated peristalsis from one region of the gut to the next i.e. absent mechanical element.

Causes of adynamic intestinal obstruction:

- a. Absent peristalsis e.g. paralytic ileus
- b. Non-propulsive e.g. mesenteric vascular occlusion [MVO], or pseudo-obstruction)

Special Considerations

- *Intra-abdominal adhesions* related to prior abdominal surgery are the most common etiologic factor of small-bowel obstruction.
- In contrast to colonic obstruction, SBO is uncommonly caused by neoplasm. Cancer-related SBO is more commonly caused by extrinsic compression or invasion by advanced malignancies arising in organs other than the small bowel.

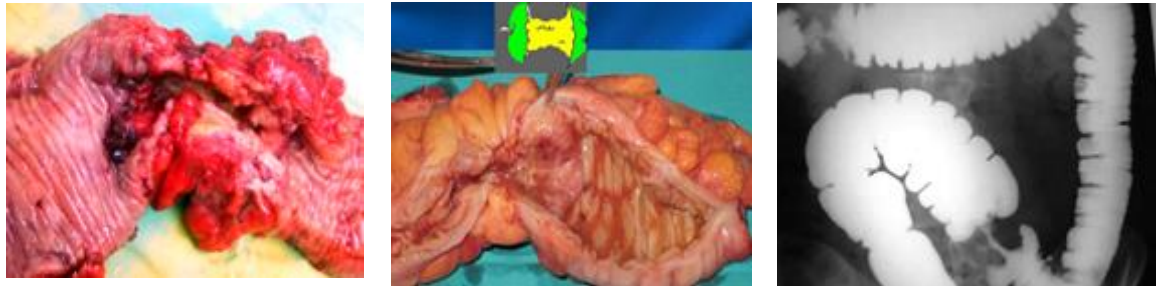


Figure 1. Obstructing colonic cancer with a classical apple-core appearance on barium enema



Figure 2. Crohn's disease of small bowel

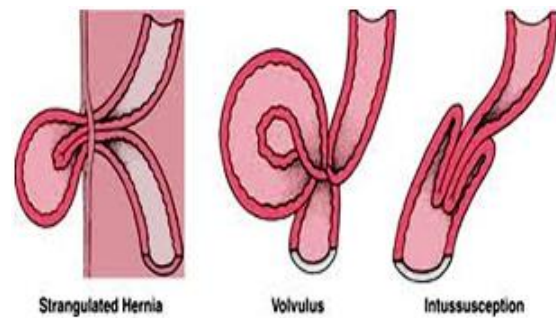


Figure 3. Commonest causes of extra-luminal obstruction

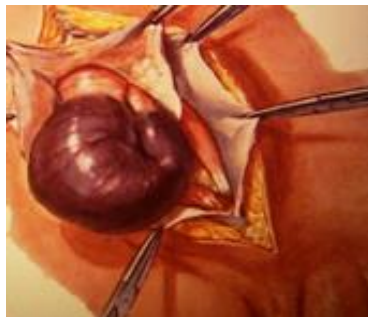


Figure 4. Strangulated inguinal hernia (pre-operatively, it is diagnosed by the 4 local cardinal signs of strangulation i.e. the hernia becomes tense, tender, irreducible & with no impulse on cough)



Figure 5. Sigmoid volvulus with classical appearance on X-ray & intra-operatively



Figure 6. GIST of the jejunum causing obstruction by jejuno-ileal Intussusception

- *Superior mesenteric artery (SMA) syndrome*; a rare etiology of obstruction caused by compression of the 3rd part of duodenum by the SMA as it crosses over it.
- *Closed loop obstruction*: The bowel is obstructed at both, proximal and distal points e.g. volvulus and tight carcinomatous stricture of the colon with a competent ileo-cecal valve (Figure 7).



Figure 7. Closed loop obstruction: tight carcinoma of the sigmoid colon with a competent ileo-cecal valve causing perforation of the hepatic flexure and transverse colon. The patient underwent total colectomy (as shown) with ileo-rectal anastomosis

PATHOPHYSIOLOGY

- Simply as blockage occurs, gas and air distend the bowel proximal (closest) to the blockage causing distension of the proximal segment leading to distension, pain and vomiting.
- Collapse of the distal segment will occur as a result of defecation and/or diagnostic enema leading to constipation that would be absolute to stools & flatus. As the process continues, gastric, bilious (bile from the liver) and pancreatic secretions begin to form a pool. Water, electrolytes & proteins accumulate in the area.
- *Distension is produced by gas and fluids*
 1. *Gas*: Swallowed + fermented + diffused from the blood).
 2. *Fluids*: Saliva (1-1.5L), gastric (1.5-2.5L), bile 1L, pancreatic 1 ½ L, small bowel (3 L).
- With ongoing gas and fluid accumulation, the bowel distends and intra-luminal and intra-mural pressure ↑. If the intra-mural pressure becomes high enough, microvascular perfusion to the intestine is impaired → intestinal ischemia. This condition is termed *strangulating bowel obstruction*. Strangulation of a bowel segment may cause necrosis (death of the tissue), perforation and loss of fluid & blood

- Since intestinal contents cannot go downstream from the stomach, nausea and vomiting occur in most patients. If the obstruction has persisted for too long or the bowel has been significantly damaged, bowel sounds decrease, eventually becoming silent.

Presentations

1. Acute obstruction
2. Chronic obstruction
3. Acute-on-chronic obstruction
4. Subacute obstruction (incomplete obstruction)

Obstruction may be:

1. *Simple*: intact blood supply
2. Strangulating/strangulated: direct interference to blood flow

Strangulation may be due to:

1. *External compression*; hernial orifices/ adhesions/ bands
2. *Interruption of mesenteric flow*: Volvulus or intussusception
3. $\uparrow$ *intra-luminal pressure*: closed-loop obstruction
4. *1^{ry} obstruction of intestinal circulation*: mesenteric infarction

What is Ileus?

It is a functional obstruction of the bowel due to lack of propulsive peristalsis of the bowel stops the movement of bowel contents downward. *Causes of ileus* include:

1. Anesthesia,
2. Interruption of nerve supply to the bowel,
3. Intestinal ischemia.
4. Abdominal wound infections, intra-abdominal sepsis
5. Electrolyte imbalance (loss of potassium $\rightarrow$ lack of intestinal peristalsis)
6. Metabolic diseases.

CLINICAL PICTURE & DIAGNOSIS

Dynamic Obstruction

1. Pain: Colicky abdominal pain.
2. Nausea and vomiting
3. Abdominal distension
4. Absolute constipation: Passage of flatus and/or stool up to 6 hours after presentation may occur and does *not* exclude intestinal obstruction
5. Bowel sounds: are classically high-pitched and active, coming on in rushes coincident with cramp pain

These features vary according to

1. Location of the obstruction: distension is most pronounced if the site of obstruction is in the distal ileum, may be absent if the site of obstruction is in the proximal jejunum.
2. Duration of the obstruction
3. Underlying pathology
4. Presence or absence of intestinal ischemia: strangulated obstruction with ischemia $\rightarrow$ tachycardia, localized abdominal tenderness, fever, marked leukocytosis, & acidosis.

Other manifestations of bowel obstruction

1. Dehydration & hypokalemia.
2. Abdominal tenderness
3. Pyrexia & mild leukocytosis

Diagnosis

Tools of Diagnosis

1. Clinical examination
2. Double enema test
3. Plain X-ray abdomen (*supine* abdominal films)
4. CT scan abdomen to diagnose obstruction and reveal its etiology.
5. Abdominal US
6. Upper and lower GI series (Gastrografin enema and Gastrografin follow-through).
Gastrografin is used as the contrast material because it is water-soluble (Barium studies are *not* recommended).

The Strategy of Diagnosis

The aim of work-up for diagnosis is to:

1. Distinguish mechanical obstruction from ileus
2. Determine etiology of obstruction
3. Discriminate partial from complete obstruction
4. Discriminate simple from strangulating obstruction

Features of Supine Abdominal Films (Figure 8)

- *Small bowel obstruction*: triad of (1) dilated small bowel loops > 3 cm in diameter, (2) air-fluid levels that are generally central & lie transversally in a step-ladder fashion across the abdomen & (3) no air in the colon.
- *Jejunum*: characterized by its valvulae conniventes, which completely pass across the width of the bowel and are regularly spaced giving a concertina effect.
- *Ileum*: featureless
- *Cecum*: rounded gas shadow in the right iliac fossa
- *Large bowel*: haustral folds which, unlike valvulae conniventes, are spaced irregularly and the indentations are *not* placed opposite one another. They do *not* usually traverse the complete width of the bowel



Figure 8. Plain X-ray abdomen showing Multiple fluid levels in SBO

TREATMENT OF ACUTE INTESTINAL OBSTRUCTION

Management Strategy

1. Gastric decompression using a nasogastric (NG) tube to reduce nausea, distension and the risk of vomiting and aspiration

2. Fluid & electrolyte resuscitation
3. Urinary catheter insertion to calculate the urine output (UOP)
4. Blood transfusion and/or blood products (FFP), if needed
5. Antibiotics as bacterial translocation may occur in the setting of bowel obstruction
6. Relief of obstruction, usually *surgical*.

Surgical Treatment

When to operate ?

- Not too early and not too late
- *Indications of early surgical intervention*
 1. Obstructed or strangulated external hernia
 2. Internal intestinal strangulation
 3. Acute obstruction

Aim of surgery - Why operate ?

- The goal is to operate before the onset of irreversible ischemia

How to operate ? What to do ?

- *Operative assessment is directed to*
 1. The *site* of obstruction
 2. The *nature* of the obstruction
 3. The viability of the gut: non-viable bowel is resected. Criteria suggesting viability are normal color, peristalsis & marginal arterial pulsations.
- *Exploration*
 1. Incision: Midline/Right paramedian
 2. Palpate the cecum: *Distended cecum = LBO, while collapsed cecum = SBO*
- Procedure – dealing with the cause
 1. Adhesions → Division (lysis)
 2. Bands → Release (cut)
 3. Hernias → Reduction & repair
 4. Tumors → Resection or bypass or diverting stoma (ileostomy – colostomy)

Specific Causes of Obstruction

Large Bowel Obstruction (LBO)

1. Obstructing lesion in the cecum or ascending colon:

- *Removable* (resectable) → emergency right hemicolectomy.
- *Irremovable* (irresectable) → proximal *stoma* (ileostomy) or ileo-transverse *bypass*.

2. Obstructing lesion in the left colon → left hemicolectomy

3. Obstructing recto-sigmoid lesions

- Resection anastomosis ± on-table colonic-lavage, ± a covering stoma
- Total colectomy with ileo-rectal anastomosis, in patients with multiple colonic carcinomas or closed-loop obstruction with devitalized or perforated cecum.
- Stenting followed later by resection anastomosis

- Colostomy and mucous fistula: The proximal colon is brought to the surface as a colostomy and the distal bowel as mucous fistula, to facilitate subsequent extra-peritoneal closure
- Hartmann`s procedure: End-colostomy closure of the rectal stump. A 2nd-stage colo-rectal anastomosis can be planned when the patient is fit.
- Colostomy: Later on, resection anastomosis is performed and finally closure of stoma in patients unfit for surgery at time of presentation

4. Rectal cancer

- *Anterior resection* (resection of the rectum through an abdominal approach)
 - o High anterior resection: Resection of distal sigmoid colon and upper rectum
 - o Low anterior resection: For lesions in the upper & mid rectum
 - o Extended low anterior resection: For lesions in the distal rectum
- *Abdomino-perineal resection (APR)*
 - o Excision of the rectum, anal canal & anus with a permanent colostomy

Obstruction by Adhesions and Bands

Etiology of adhesive intestinal obstruction

1. Operative injury
2. Bacterial peritonitis
3. Radiotherapy
4. Ischemic injury
5. Chemical injury
6. FB reaction

Problems of adhesive intestinal obstruction

1. Small intestinal obstruction (SBO)
2. 2^{ry} female infertility
3. Ectopic gestation
4. Chronic abdominal and pelvic pain
5. Hazardous re-operation

Factors that limit adhesion formation

1. Good surgical techniques
2. Minimize contact with gauze.
3. Cover anastomosis and raw peritoneal surfaces
4. Seprafilm: Hyaluronic acid and Carboxy-methyl-cellulose

Bands

1. *Congenital*: e.g. obliterated vitello-intestinal duct.
2. *Acquired*: A string band following previous bacterial peritonitis.

Treatment

- NG tube decompression & IV rehydration.
- Laparotomy, if needed, to divide causative adhesions.
- R/ of recurrent intestinal obstruction due to adhesions:
 1. Repeat adhesiolysis alone (open / laparoscopic)



Figure 9. Small bowel obstruction 2^{ry} to intra-peritoneal fibrous band adhesion

2. Adhesiolysis with seprafilm
 3. *Noble's* intestinal plication (Figure 10)
 4. *Child-Phillips* trans-mesenteric plication (Figure 11).
 5. *Baker* intestinal intubation (Figure 12).
- Their relative efficacy remains unclear

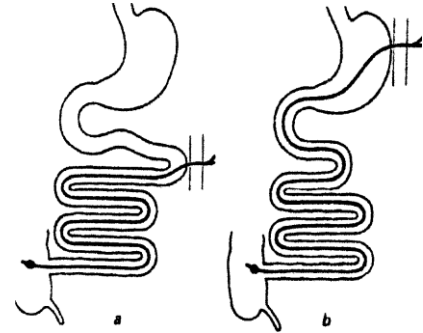
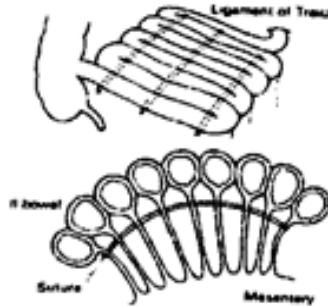
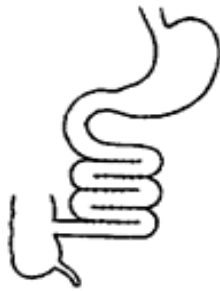


Figure 10. Noble's plication

Figure 11. Child-Phillips plication

Figure 12. Baker's intubation

Internal Hernia

Definition

- Portion of the small intestine becomes entrapped in one of the retroperitoneal fossae or into a congenital mesenteric defect.

Potential sites of internal herniation

- The foramen of Winslow.
- A hole in the mesentery.
- A hole in the transverse mesocolon.
- Defects in the broad ligament.
- Congenital or acquired diaphragmatic hernia.
- Duodenal retro-peritoneal fossae, left para-duodenal & right duodeno-jejunal.
- Cecal/appendiceal retro-peritoneal fossae - superior, inferior & retro-cecal.
- Inter-sigmoid fossa.

Treatment of internal herniation

- The standard R/ is to release the constricting agent by division
- This should not be undertaken in cases of herniation involving the foramen of Winslow, mesenteric defects & the para-duodenal / duodeno-jejunal fossae, as major blood vessels run in the edge of the constriction ring. The distended loop in such circumstances must first be decompressed with minimal contamination & then reduced.

Enteric Stricture Causing SBO

- *Inflammatory* stricture 2ry to TB or Crohn's disease
- *Malignant* stricture: lymphoma is common, whilst carcinoma and sarcoma are rare.
- *Presentation* is usually *subacute* or *chronic*
- *Standard surgical R/*: Resection & anastomosis. In Crohn's disease *stricturoplasty* may be considered in the presence of short multiple strictures without active sepsis.

Bolus Obstruction

- Food
- Tricho-bezoars & phyto-bezoars
- Stercoliths
- *Worms*: *Ascaris lumbricoides*
- *Gallstones*: in the elderly, 2^{ry} to erosion of a large gallstone through the GB into the duodenum. Classically, there is impaction about 60 cm proximal to the ileo-cecal valve (Figure 13).



Figure 13. Impaction of a gallstone in the ileum (60 cm proximal to the ileo-cecal valve).

Gastrointestinal Obstruction in Infancy & Childhood

1. Infantile hypertrophic pyloric stenosis
2. Malrotation & volvulus neonatorum
3. Anorectal malformations
4. Intussusception
5. Intestinal atresia
6. Meconium ileus
7. Hirschsprung`s disease (Aganglionic megacolon)

CHRONIC INTESTINAL OBSTRUCITON

Organic

- *Intra-mural*: fecal impaction
- *Mural*: colorectal cancer, diverticulitis, strictures (Crohn`s disease, ischemia), anastomotic stenosis.
- *Extra-mural*: adhesion (small bowel only), metastatic deposits, endometriosis

Functional

- Hirschprung`s disease, idiopathic megacolon, pseudo-obstruction

ADYNAMIC OBSTRUCITON

Ileus and pseudo-obstruction designate clinical syndromes caused by impaired intestinal motility and are characterized by symptoms and signs of intestinal obstruction in the absence of a lesion-causing mechanical obstruction.

Paralytic ileus

Varieties (Causes)

1. Post-operative
2. Infection: Intra-abdominal sepsis, abscess, peritonitis
3. Reflex ileus: following fractures of the spine or ribs, retro-peritoneal hemorrhage.
4. Metabolic: uremia, hypokalemia, hypo-thyroidism
5. Medications: anticholinergics, opiates, calcium channel blockers

Clinical Features

- Distension
- Effortless vomiting
- Absence of bowel sounds
- No passage of flatus
- Pain is not a feature
- X-ray: gas-filled intestinal loops with multiple fluid levels
- CT scan is the test of choice

Management of Paralytic Ileus

- Nasogastric suction
- Fluid and electrolyte balance
- Removal of the 1st cause
- No place for routine use of peristaltic stimulants.
- If paralytic ileus is prolonged; *laparotomy* is performed to exclude a hidden cause and facilitate bowel decompression

Pseudo-obstruction

- *Definition*: Obstruction, usually of the colon, in the absence of a mechanical cause or acute intra-abdominal disease. It is associated with a variety of syndromes where there is an underlying neuropathy &/or myopathy.
- *Small intestinal pseudo-obstruction*
 1. Primary (i.e. sporadic or familial visceral myopathy, neuropathy).
 2. Secondary (e.g. scleroderma, muscular dystrophy, spinal cord injury, viral infection)
- *Colonic pseudo-obstruction*
 1. Acute form: **Ogilvie syndrome**
 2. Chronic form

Mesenteric Ischemia

Causes of acute mesenteric ischemia

- Arterial or venous thrombosis – Embolism – Vasculitis - Surgical trauma to vessels (e.g. aortic reconstruction)

Gut ischemia: important anatomical consideration

- Marginal artery of *Drummond* - Marginal artery of *Dwight - Griffiths'* point - The arc of *Riolan*

Acute Mesenteric Ischemia

Clinical features and diagnosis

- Sudden onset of severe abdominal pain in a patient with atrial fibrillation or atherosclerosis. Abdominal pain is classically out of proportion to the physical findings
- Persistent vomiting and defecation occur early with subsequent passage of altered blood.
- Hypo-volemic shock rapidly ensues.
- Mild abdominal tenderness initially. Rigidity is a late feature.
- 40% of patients receive no treatment or an “open & close” laparotomy
- *Laboratory tests:* Leucocytosis, acidosis, hyper-amylasemia
- *Imaging:* Plain X-ray, angiography, CT scan.

Treatment

- Full resuscitation
- Massive resection of affected bowel (Figure 14) unless flow is restored within 6 hours
- Second look laparotomy
- In early cases: embolectomy or revascularisation of the SMA by aorto-SMA bypass.
- Anti-coagulations early in the post-operative period.
- IV alimentation after extensive enterectomy.
- Small bowel transplantation in selected cases.

Prognosis

- Mortality rate = 60-85% !!



Figure 14. Extensive arterial mesenteric ischemia causing gangrene of a large segment of bowel (resected during operation)

Chronic Mesenteric Ischemia (Intestinal Angina)

- *Cause:* insufficient blood flow to the bowel during periods of the increased metabolic demand associated with digestion.
- *Symptoms:* Recurrent postprandial pain, bloating, flatulence, constipation, or diarrhea, associated with weight loss and steatorrhea (50%).
- Patients are usually in their 50s or 60s and have other evidence of generalized atherosclerosis.

Upper Gastrointestinal (UGI) Bleeding

Definition

Upper gastrointestinal (UGI) bleeding means bleeding from the gastro-intestinal tract (GIT) proximal to the ligament of *Treitz*. Bleeding site is no more distal to the duodenum.

CAUSES OF UGI BLEEDING

Causes of UGI result from the esophagus, stomach or duodenum (Table 1).

Table 1: Etiology of UGI bleeding

Esophageal Causes	Gastric Causes	Duodenal Causes
- Esophageal varices - Hiatal hernia - Reflux esophagitis - Esophageal carcinoma - Mallory Weiss tear	- Gastric peptic ulcer - Stress ulcer - Erosive gastritis - Portal hypertensive gastropathy - Gastric varices - Dieulafoy lesion - Menetrier's syndrome - Gastric tumors (carcinoma, lymphoma, GIST) - Rupture of retro-gastric aortic aneurysm (aorto-enteric fistula).	- Duodenal peptic ulcer - Duodenal diverticulum - Duodenitis - Carcinoma of duodenum - Lymphoma - Hemobilia - Crohn's disease - Duodenal diverticulum

GIST: Gastro-intestinal stromal tumor

Most common causes in Egypt (in order of frequency)

1. Esophageal varices (75%)
2. Erosive gastritis
3. Duodenal ulcer (DU)
4. Cancer stomach
5. Gastric ulcer (GU)

CLINICAL PRESENTATION

1. Hematemesis
 - Fresh blood (significant bleeding)
 - Small amount of altered blood (coffee grounds) = mild bleeding that already settled.
2. Melena
 - Liquid, jet black or black with reddish tinge pungent characteristic smell, quite unlike smell of feces.
 - DD from Iron therapy (sticky feces with dark grey rather than black)
 - Massive bleeding may cause fresh rectal bleeding or large fresh clots per rectum.
3. Both hematemesis & melena

The severity of the attack can be evaluated by the presence & extent of hypovolemia.

1. *Mild*: no significant hypovolemia.
2. *Moderate*: hypovolemia that responds to volume replacement (crystalloids & blood) & thereafter the patient is stable.
3. *Severe*: active continued major bleeding rendering resuscitation with blood transfusion difficult, or recurrent major re-bleeding after successful resuscitation.

Presentation according to onset

1. Sudden massive hematemesis → shock
2. Slow (usually melena) → chronic anemia

TREATMENT OF ACUTE UGI BLEEDING

Resuscitation

- All patients should be kept *nothing by mouth* at least until endoscopy has been performed.
- *Fluid replacement & blood transfusion*
 - o Send blood for grouping & save, start slow IV.
 - o Blood transfusion if Hgb is <10 g/dl with evidence of acute blood loss.
 - o Goal of a hematocrit value (HCV) > 30%.
 - o Rapid transfusion in hemodynamically stable patient should be avoided (Heart failure and re-bleeding).
 - o Warm the blood to 25°C by warmer.
 - o Fresh frozen plasma (FFP) every 6 units of stored blood (15 ml/kg) or every 4 U packed RBCs.
 - o Citrate toxicity is avoided by 10 ml of 10% Ca gluconate after the first 6 units then every 2 subsequent units. Hypokalemia is carefully monitored.

Surgical Treatment

Indications of Surgery

1. Patients with severe hemorrhage that resuscitation with volume replacement cannot keep up with the losses.

2. Elderly patients older than 60 y if > 4 units of blood are necessary during the initial resuscitation and patient who has one recurrence of bleeding after initial successful control

Lines of Surgery

1. The use of a definitive ulcer-curing operation is mandatory in patients with hemorrhage, but is optional in patients with perforation.
2. Pyloromyotomy under run the vessel.
3. PG or TV + GJ (truncal vagotomy and gastro-jejunostomy)
4. Recurrent bleeding in both is 13%
5. Mortality of PG 8% is double that of TV + GJ.

Treatment of Bleeding Peptic Ulcer Disease

Bleeding occur in 20% of patients with DU. It may occur without history of dyspepsia.

Diagnosis

- Typical history
- Epigastric tenderness
- Endoscopy: not only diagnosis
 1. Site: posterior.
 2. Risk signs:
 - a. Visible vessel (blue or red) 50% risk of re-bleeding.
 - b. Spurter (arterial) 50% risk of re-bleeding
 - c. Overlying adherent clot 10% risk of re-bleeding
 - d. Red or black spot in the base of the ulcer 10% risk of re-bleeding

Treatment

Approximately, 80% will stop without intervention, so any technique should be subjected to controlled trial.

A. *Medical treatment*: H₂ blockers, proton pump inhibitors, somatostatin, prostaglandins.

B. *Endoscopic techniques for hemostasis*:

1. Injection methods
 - Adrenaline (1:10,000): Tissue pressure, active vasospasm, activation of platelets
 - Sclerosants as STD, polidocanol act by damaging the endothelium.
 - Absolute alcohol: rapid dehydration ... tissue fixation.
2. Laser photocoagulation: Argon, Nd YAG.
3. Diathermy, bipolar
4. Heater probe
5. Topical agent: Collagen, clotting factors, cyanoacrylate glue, fibrin sealant, thrombin.
6. Mechanical methods: *Clips, staples, sutures*.
7. Microwave and argon plasma coagulator: (*under study*).

NB. The endoscopic techniques for visible vessels and active bleeding are:

- Thermal
- Electrocoagulation
- Photocoagulation
- Injection therapy

C. Insertion of NG tube, lavage with ice cold saline to induce gastric mucosal constriction.

D. Pharmacological control of bleeding: H₂-blockers, sucralfate, proton-pump inhibitors, vasopressin (or analogues) and somatostatin. No drug has appreciable effect on ulcer

E. Vascular embolization.

F. Surgical treatment

- With the expansion and increased efficacy of endoscopic techniques, fewer patients are treated surgically nowadays.
- Indications for surgery (in about 25%)
 1. Massive bleeding requiring >10 units.
 2. Rebleeding after cessation (Death rate >30%) so need Early surgery
 3. Visible bleeding vessel on endoscopy (50% rebleeding).
- Relative indications: giant DU or GU, visible vessel, shortage of blood and previous bleeding.

G. With stopping of bleeding, after 12-24 h the patient can eat to ↑ the morale, stimulate muscle contraction and ↓ acidity Order hematocrit and occult blood in stools (twice daily).

Rebleeding PU

- Incidence: 5%
- Most patients re-bleed within 2 days from the time of first episode
- Re-bleeding carries tenfold increase in mortality (MR = 25%).
- Risk signs carry a 40% risk of re-bleeding compared to only 4% in absence of risk signs.
- If risk signs are present → active treatment, if re-bleeding → surgery.

Prognosis

- MR of emergency operation is 8-10% versus 1-2% in elective or semi-elective surgery

Treatment Options for Erosive Gastritis

1. Near total gastrotomy + Roux-en-Y anastomosis.
2. Ligation of all blood vessels to the stomach.
3. Vagotomy and pyloroplasty.
4. Conserve until transfusion requirements are 12 units or more.

Treatment of Esophageal Varices

General Plan

- Treatment of shock and guard against liver failure and encephalopathy
- Stop bleeding by chemical, mechanical, physical or surgical methods.

Pharmacological Treatment

- *Vasopressin*: IV bolus: 20 units / 100 ml G 5% / 20 min or Infusion 0.4 units / min. It has 3 side effects; myocardial infarction, gut ischemia & local gangrene
- *Glypressin*: It has lower side effects. Bolus dose, longer action

- *Somatostatin*: Synthetic analogue (Octereotide), half-life 1-2 hours, greater potency, little side effects. Dose = mg/8 h for 48 hours on 250 ml of glucose 5%

Surgical Treatment

Indications for surgery

- Non stoppage of bleeding after 2 sessions of sclerotherapy
- Bleeding gastric varices (5-50% of cases)

Operative lines

- Emergency decongestion: Mortality = 40%. Portal vein thrombosis occurs in 30% of cases
- Emergency porto-caval shunt is *obsolete*
- The preferred technique is esophageal transection with stapling.

Differential Diagnosis of Abdominal Masses

ABDOMINAL EXAMINATION

Exposure

- From the nipple to the knee,
- If embarrassing, cover the lower abdomen with a sheet.

Inspection

Abdominal Contour

- *Normal contour*: The abdomen is flat from the xiphoid to the pubis and the umbilicus is at the center of the abdomen.
- *Generalized distension* (Figure 1): Due to fat, fetus, feces, flatus, fluid, full-sized tumors
- *Localized bulge* (Figure 2): Due to mass, organomegaly, hernia.



Figure 1: Generalized abdominal distention

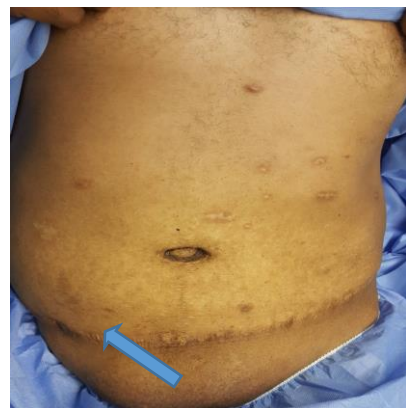


Figure 2: Localized distention (arrow)

Palpation

Superficial Palpation

- To gain the patient's confidence
- To detect local temperature, tenderness, parietal mass, or hypersthesia

Deep Palpation

- To feel the following:
 1. Abdominal organs: Liver, spleen, kidneys
 2. Abdominal aorta
 3. Any masses
- If masses are felt, note the following:
 1. 6S: Site, size, shape, surface, skin overlying, special character:
 2. Consistency
 3. Tenderness
 4. Pulsations
 5. Mobility with respiration or with hand.

Parietal versus Intra-Abdominal Mass

The *rising-up test*, with the aid of *imaging*, can differentiate between parietal and intra-abdominal masses.

Parietal Swellings

- Extends over the costal margin
- Moves anterior and posterior with respiration
- More prominent on rising-up test

Intra-abdominal Swellings

- Disappears beneath costal margin
- Moves up and down with respiration
- Less prominent on rising-up test

1. COMMON PARIETAL SWELLINGS

Skin

- Sebaceous cyst
- Papilloma
- Melanoma
- Squamous cell carcinoma (SCC)

Subcutaneous Tissue

- Lipoma: Subcutaneous (SC), inter-muscular
- Neurofibroma
- Hemangioma
- Lymphangioma

Muscles

- Rectus sheath hematoma
- Desmoid tumor

ABDOMINAL QUADRANTS / REGIONS

Nine Regions

The abdomen is divided by 2 vertical planes and 2 transverse planes into 9 regions:

- The 2 vertical planes are mid-clavicular **or** pass from the mid-inguinal point upwards, on each side.
- The 2 transverse planes:
 1. The upper is roughly midway between the xiphoid and umbilicus, or touching the lower costal margin (subcostal line), or at the trans-pyloric plane of Addison", which passes at the lower edge of L1.
 2. The lower plane is roughly midway between the umbilicus and symphysis pubis, or between the highest points of the iliac crests (inter-costal line) or between the tubercles of the iliac crest (inter-tubercular line).

Accordingly, the 9 regions are as follows (Figure 3):

1. Right hypochondrium.
2. Epigastrium
3. Left hypochondrium.
4. Right lumbar region.
5. Umbilical region.
6. Left lumbar region.
7. Right iliac fossa (RIF).
8. Hypogastrium (supra-umbilical region)
9. Left iliac fossa (LIF).

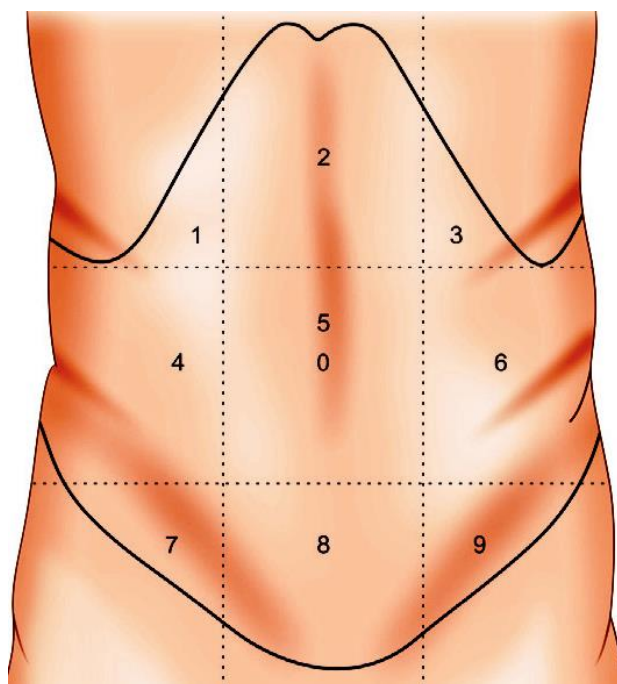


Figure 3. The nine regions of the abdomen

Four Quadrants

The abdomen can also be divided into 4 quadrants by a midline *vertical plane* passing from the xiphoid to the symphysis pubis and a *transverse plane* passing horizontally through the umbilicus.

Accordingly the 4 quadrants of the abdomen are:

1. Upper right quadrant (URQ)
2. Upper left quadrant (ULQ)
3. Lower right quadrant (RLQ)
4. Lower left quadrant (LLQ)

Figure 4 demonstrates the 9 regions of the abdomen (a) and the 4 quadrants of the abdomen (b)

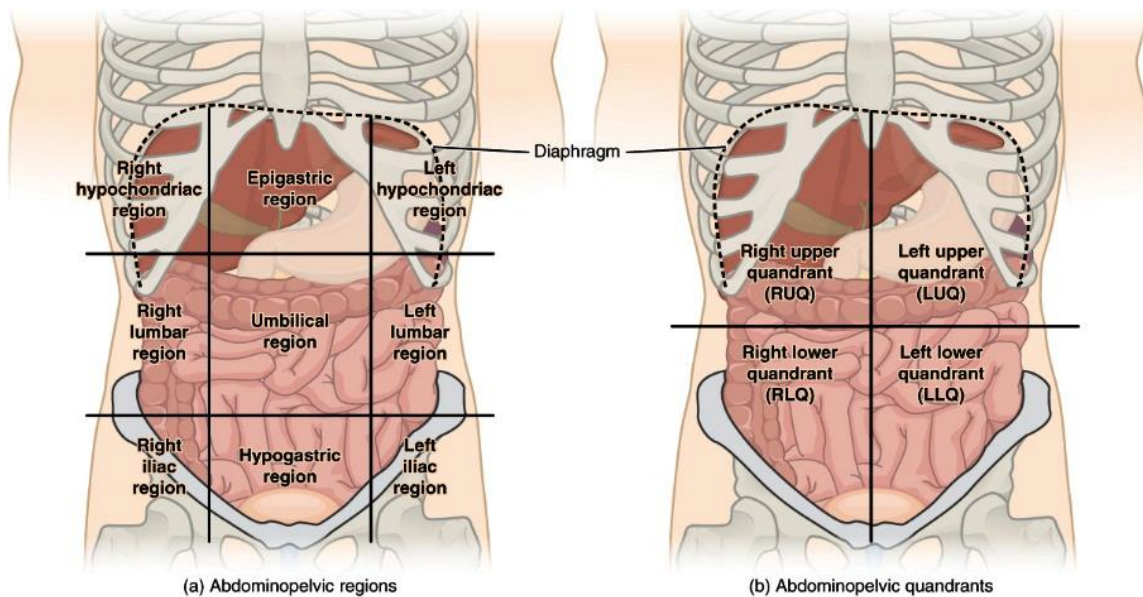


Figure 4: The 9 abdomino-pelvic regions (a) and 4 abdomino-pelvic quadrants (b)

Figure 5 demonstrates the specific organs occupying the different 9 regions of the abdomen, which aids the diagnosis of a ‘mass# occupying a certain region

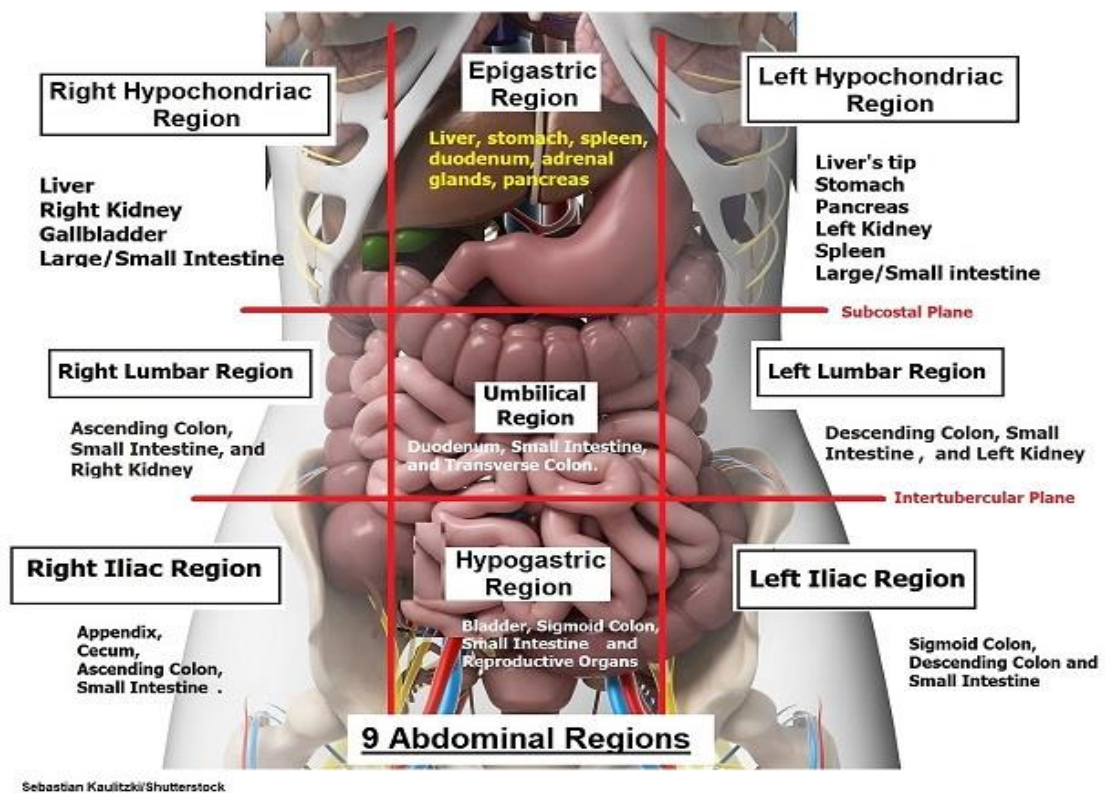


Figure 5. The different organs occupying the 9 regions of the abdomen

CLINICAL APPROACH

In a patient presenting with an abdominal mass, the following clinical features, should be carefully assessed:

- *Pain*: Site, nature, aggravating or relieving factors, duration of pain, referred pain.
- *Vomiting*: Type, content, hematemesis, relation to food, frequency.
- *Jaundice*: It is an important factor in relation to liver, gall bladder or pancreatic masses.
- *Bowel habits*: Constipation, diarrhea, bloody diarrhea, furious diarrhea, tenesmus.
- *Decreased appetite and weight*.
- *Inspection of the mass*: Anatomical location, margin, movement with respiration.
- *Palpation of the mass*: Site, extent, surface, tenderness, consistency, movement with respiration, mobility, borders, plane of the swelling (by leg-raising test), presence of other masses.
- *Often the mass needs to be examined for change of position*—in sitting, in standing, in side position, after a brisk walk, in knee-elbow position for retro-peritoneal mass and *for puddle sign* (but difficult to keep patient in this position).
- *Percussion* is an important aspect of examination in case of an abdominal mass. Percussion over the mass is important to predict the anatomical location of the mass.
 1. If the mass has a dull note, then it is in the anterior abdominal wall, or in front of the bowel intra-abdominally like liver, spleen, gall bladder (GB).
 2. If the mass is with an impaired resonant note, then the mass is arising from the bowel like stomach, colon, or small bowel.
 3. If the mass is resonant on percussion, then it is probably in the retro-peritoneal region.
 4. Other than this, liver dullness, free fluid in the abdomen should be elicited during percussion.
- *Per-rectal (PR) examination*:
It is done to look for any secondaries in recto-vesical pouch, primary tumor or relation of lower abdomen masses (pelvic masses).
- *Per-vaginal (PV) examination*:
It is done to assess pelvic masses.

MASS IN THE RIGHT HYPOCHONDRIMUM

A. Palpable Liver Mass in the Right Hypochondrium

Characteristics

- It is horizontally placed.
- It usually moves with respiration.
- The upper border is not felt.
- It is dull on percussion (this dullness continues over liver dullness above).
- Fingers cannot be insinuated under the right costal margin.

Causes of an Enlarged Liver

1. *Soft, Smooth, Non-Tender Liver*

- *Hydro-hepatosis*: It is due to obstruction of the common bile duct (CBD) causing dilatation of the intra-hepatic biliary radicles.
- Congestive cardiac failure.
- *Hydatid cyst of the liver*: The mass is well-localized in the liver with a typical hydatid thrill. Three fingers are placed over the mass widely; when the central finger is tapped, fluid movement is elicited in lateral two fingers (*Three finger test*).

2. *Soft, Smooth, Tender Liver*

Amoebic liver abscess

- It is due to entameba histolytica infestation
- Common in males (20:1), fever, pain, intercostal tenderness, tender liver
- It is more common in alcoholics and cirrhotics
- Single abscess is common (70%), most common in the right postero-superior lobe (80%)
- The liver often gets adherent to the anterior abdominal wall and will not move with respiration. Inter-costal tenderness and right-sided pleural effusion are common.
- Chocolate-colored Anchovy sauce pus is classical
- Secondary infection can occur (30%) and is life-threatening due to septicemia
- It can be acute or chronic; both mimics hepatoma
- The most common site of rupture is into the lungs.
- The most dangerous rupture is into pericardium (left lobe abscess)
- Liver failure can develop in cirrhotic patients
- Mimics cholecystitis, subphrenic abscess, hepatoma
- Total count, LFT, prothrombin time, US abdomen are relevant investigations
- Chest X-ray may show left-sided sympathetic pleural effusion
- CT scan to differentiate it from hepatoma
- *Treatment*: Drugs like metronidazole, injection dehydroemetine, chloroquine tablets, diloxanate furoate; US-guided aspiration after controlling prothrombin time using vitamin K or fresh frozen plasma (FFP). If it recurs, percutaneous guided drainage using a pigtail catheter, or open laparotomy and drainage with placement of Malecot's catheter may be used.

3. *Hard, Smooth Liver (Hepatic Tumors)*

- *Hepatoma (HCC)*
- Solitary secondary in liver.

Hepatoma / hepato-cellular carcinoma (HCC)

- Common etiologies are afla toxins, hepatitis B and C virus infection, alcoholic cirrhosis, hemochromatosis, smoking, hepatic adenoma, clonorchis sinensis, polyvinyl chloride
- Uni-centric tumor and right lobe involvement is more common

- Fibro-lamellar variant is common in the left lobe, not related to hepatitis or cirrhosis and without alpha-feto protein (AFP) level raise. There are increased serum vitamin B₁₂ binding capacity and neurotensin levels.
- It can be multifocal/indeterminate/spreading/expanding— *Okuda classification*
- Presents as a large smooth hard liver mass — later jaundice, fever, pain and tenderness, ascites and bruit over the mass. Occasionally there can be multiple nodules when it is multi-centric.
- A rapidly growing tumor can be also soft.
- Hepatoma often can also be tender due to tumor necrosis or stretching of the liver capsule.
- A vascular bruit may be heard over the liver during auscultation.
- Spreads to lymphatics, blood and direct spread
- Mimics amebic liver abscess, secondaries, hydatid cyst, polycystic liver disease
- LFT, CT scan, raised AFP, liver biopsy (only needed) are the investigations
- Hemi-hepatectomy in early operable growth is the treatment
- Hepatic artery ligation / intra-arterial chemotherapy /chemoembolization / percutaneous ethanol or acetic acid injection / radiofrequency ablation/chemotherapy using adriamycin, carboplatin, gemcitabine — are palliative procedures

4. **Hard, Multi-Nodular Liver**

- **Multiple secondaries in liver**
Here, hard nodules show *umbilication* due to central necrosis.

B. Palpable Gall bladder (GB) in the Right Hypochondrium

Characteristics

- It is smooth and soft (except in carcinoma of the GB).
- It is mobile horizontally (side-to-side).
- It moves with respiration.
- It is located right of the right rectus muscle, below the right costal margin or below the lower margin of the palpable liver.
- It is dull on percussion.

Causes of a Palpable Gall Bladder

1. *Soft, non-tender gall bladder*
 - Mucocele of the gall bladder.
 - Enlarged gall bladder in obstructive jaundice due to carcinoma head of the pancreas or peri ampullary carcinoma or growth in the CBD.
2. *Hard gall bladder:* Carcinoma gall bladder
3. *Tender gall bladder:* Empyema GB

C. Other Masses in the Right Hypochondrium

1. *Peri-cholecystic inflammatory mass*: It is tender, smooth, firm or soft, non-mobile, intra-abdominal mass often with guarding.
2. *Kidney mass arising from upper pole*: due to renal cell carcinoma or hydronephrosis.

MASS IN THE EPIGASTRIUM

Palpable Left Lobe of the Liver

- It is in the epigastric region.
- Its upper border cannot be felt.
- It moves with respiration.
- It extends towards the left hypochondrium.
- It is dull on percussion.

Features of Stomach Mass

- It is located in the epigastric region.
- It is an intra-abdominal mass that moves with respiration.
- It is resonant or impaired resonant on percussion.
- The mass may be better felt on standing or on walking. It is often mobile, unless it gets adherent posteriorly. Its presentation depends on the gastric region from which it arises:
 1. *Pylorus*: all margins of the mass are well felt. It is mobile, with features of gastric outlet obstruction.
 2. *Body*: the mass is horizontally placed without any features of obstruction.
 3. *Upper part of the stomach* near the gastro-esophageal junction: it causes dysphagia.
 4. *Fundus*: the mass is in the upper part of the epigastric region towards left side.
- Gastro-intestinal stromal tumor (GIST) of stomach is smooth and firm.
- Stomach carcinoma is nodular and hard (Figure 6). It is the most common cause of stomach mass

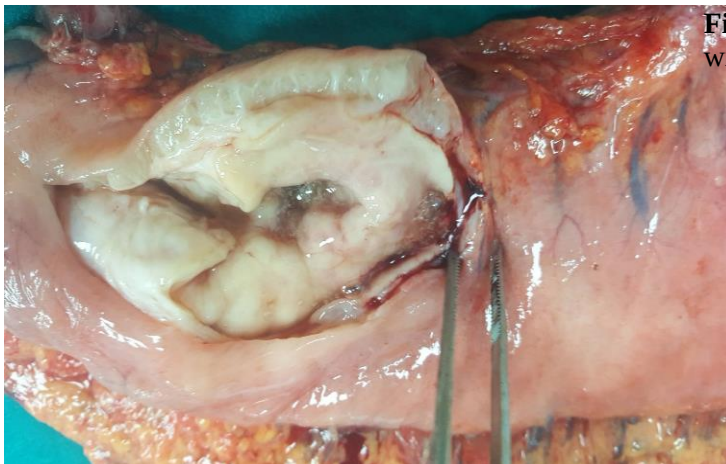


Figure 6: Gastric carcinoma present with a hard, nodular mass

- *Management of gastric carcinoma*

1. Early growth in the pylorus: lower radical gastrectomy with removal of the tumor, proximal 5 cm clearance, nodal clearance, greater and lesser omentum, distal pancreas and spleen (now not regularly removed; it is removed to clear splenic lymph nodes and Billroth II anastomosis or Roux-en-Y anastomosis is done. Post-operatively adjuvant chemotherapy should be given (5-fluorouracil, mitomycin, epirubicin, cisplatin).
2. Growth in body, proximal growth, diffuse carcinoma and generalized linitis plastica are the indications of total radical gastrectomy with esophago-jejunal anastomosis
3. Neoadjuvant chemotherapy in advanced gastric cancer prior to gastrectomy
4. Instillation of mitomycin C impregnated charcoal intra-peritoneally to control lymphatic disease (Japan)
5. Palliative procedures like palliative partial gastrectomy, anterior gastro-jejunostomy, Devine's exclusion procedure, luminal stenting in proximal inoperable growths, chemotherapy are used in inoperable cases
6. In early carcinoma, proper lymph nodal clearance is important

Abdominal Masses of Pancreatic Origin (Figure 7)

A. *Pancreatic Cystic Lesions (inflammatory cystic or cystic neoplasm)*

B. *WOPN (walled-off pancreatic necrosis),*

C. *Pseudo-cyst of the Pancreas*

- Mass in the epigastric region, smooth, soft, and may be tender if it gets infected.
- It does *not* move with respiration.
- It is *not* mobile.
- It has got transmitted pulsation.
- It is confirmed by placing the patient in knee-elbow position.
- The lower border is well felt, but the upper border is not clear.
- It is resonant on percussion.
- *Baid test*: As the stomach is pushed in front, a Ryle's tube when passed can be felt per abdomen on palpation.

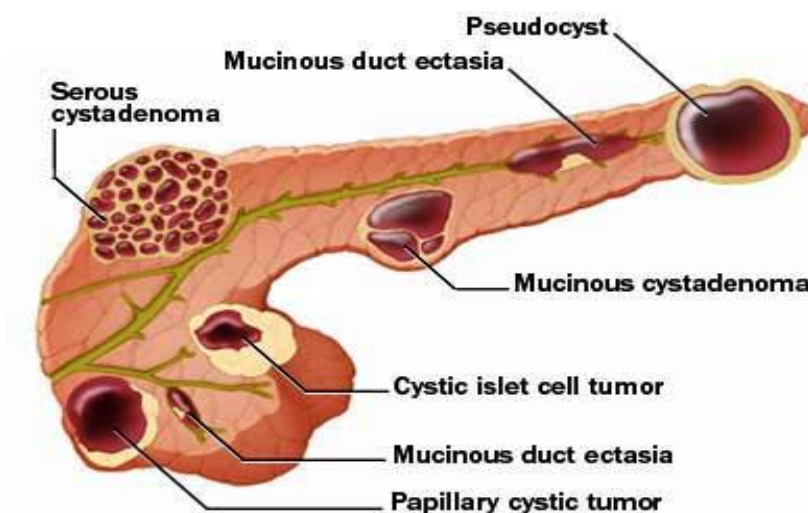


Figure 7: Different pancreatic tumors arising from different parts of the pancreas.

- *Investigations for pseudo-cyst of pancreas*
 1. Ultrasound (US): a commonly done procedure
 2. CT scan: ideal choice
 3. LFT, serum amylase, prothrombin time
 4. ERCP to find out communications
 5. Barium meal (not done nowadays): shows widened vertebro-gastric angle
- *Indications of intervention*
 1. Cyst size bigger than 6 cm
 2. Mature thick wall cyst
 3. Infected cyst
- *Complications*
 1. Rupture — 3%
 2. Infection — 20%
 3. Bleeding — torrential 7%
 4. Cholangitis
- *Interventions*
 1. Roux-en-Y cysto-jejunostomy is ideal
 2. Cysto-gastrostomy (Jurasz procedure) commonly done cysto-duodenostomy
 3. Cysto-gastrostomy with external drainage if infected (Smith operation)
 4. Endoscopic stenting
 5. Laparoscopic cysto-gastrostomy (popular and safer than open cysto-gastrostomy)
 6. Guided aspiration helps, but high recurrence rate is high (70%)

D. Pancreatic Cystic Neoplasms (Figure 8)

- Mass is smooth, firm, does not move with respiration, and non-mobile
- Resonant on percussion.
- The patient complaints of back pain.



Figure 8: Pancreatic cystic neoplasms present with a smooth, firm mass.

Serous cyst-adenomas (SCAs)

- Have thick, fibrous walls and contain clear fluid.
- Almost all SCAs are benign.

Intra-ductal Papillary Mucinous Neoplasms (IPMNs)

- Start in the ducts that connect the pancreas to the intestine.
- They are the most common type of pre-cancerous cyst.
- They produce large amounts of proteins that form mucus (mucin) in the cyst lining and fluid.
- It is difficult to predict if and when an IPMN will become cancerous, although research indicates that those that involve the main pancreatic duct are at a higher risk.

Mucinous Cystic Neoplasms (MCNs)

- These cysts are pre-cancerous growths that can start in the body and tail of the pancreas.
- They almost always develop in women rather than men.
- Large ones that contain tiny walls that divide the cyst into compartments, called septations, may be more likely to become cancerous.

E. Pancreatic Cancer (advanced cancer head pancreas, cancer body pancreas)

- It is the fourth leading cause of cancer deaths, being responsible for 8% of all cancer-related deaths.
- Approximately, 75% of all pancreatic carcinomas occur within the head or neck of the pancreas, 15-20% occurs in the body of the pancreas, and 5-10% occurs in the tail.
- Palpable GB (i.e. *Courvoisier* sign)
- Advanced intra-abdominal disease: Presence of ascites, a palpable abdominal mass, hepatomegaly from liver metastases, or splenomegaly from portal vein obstruction
- Advanced disease: Para-umbilical subcutaneous (SC) metastases (or *Sister Mary Joseph* nodule)
- Possible presence of palpable metastatic mass in the rectal pouch (*Blumer* shelf)
- Possible presence of palpable metastatic cervical lymph nodes (LNs): Nodes may be palpable behind the medial end of the left clavicle (*Virchow* node) and other areas in the cervical region

Colonic Mass

- It is due to carcinoma of transverse colon.
- It is mobile, horizontally placed, nodular, hard mass (Figure 9), which does not move with respiration.
- The cecum will be dilated and palpable.
- It is resonant or impaired resonant on percussion.
- Patient has bowel symptoms, loss of appetite and decreased weight.



Figure 9. A colonic mass is hard, nodular and horizontally placed.

Para-aortic Lymph Node Mass

- Mass in the epigastric region, which is deeply placed, non-mobile, not moving with respiration.
- It is vertically placed, above the level of the umbilicus and resonant on percussion.
- Causes of enlargement are secondaries, lymphomas or tuberculosis (TB).

Aortic Aneurysm

- It is smooth, soft, pulsatile (expansile pulsation, which is confirmed by placing the patient in knee-elbow position).
- It is vertically placed above the level of the umbilicus, non-mobile, not moving with respiration and resonant on percussion.

MASS IN THE LEFT HYPOCHONDRIUM

Enlarged Spleen

Characteristics

- Spleen has to enlarge three times to be palpable clinically.
- It enlarges from left costal margin towards the right iliac fossa.
- It moves with respiration, mobile, obliquely placed, smooth, soft or firm, with a notch on the anterior edge, which is directed downwards and inwards.
- Fingers cannot be insinuated over the upper border.
- “*Hook sign*” is positive, i.e. one cannot insinuate the fingers under the left costal margin.
- It is dull on percussion.

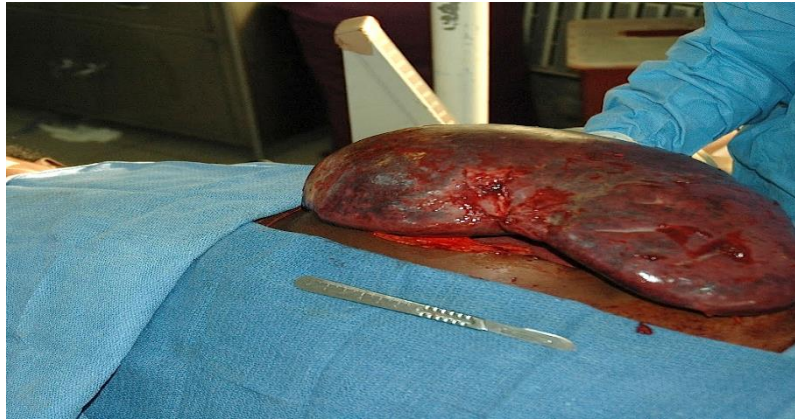


Figure 10: Splenomegaly (enlarged spleen) presents with a mass in the left hypochondrium.

Causes of splenomegaly

- *Infectious*
 1. Protozoa: Bilharziasis, malaria
 2. Viral: Cytomegalovirus (CMV), Epstein-Barr virus (EBV)
 3. Bacterial: Septicemia, TB, typhoid
- *Cellular infiltration*: Amyloidosis, sarcoidosis
- *Collagen disease*: Felty's disease
- *Space-occupying lesions*: Hydatid cyst
- *Cellular proliferative disorders*: Leukemias, lymphoma, spherocytosis, hemolytic anemia, myelo-fibrosis, sarcoidosis
- *Infarction*: Emboli, splenic artery or vein thrombosis
- *Trauma*: Splenic sub-capsular hematoma

Left-Sided Colonic Mass

- It is mobile, nodular, and resonant.
- It does *not* move with respiration.
- It is commonly due to carcinoma colon.

Left Renal Mass from Upper Pole of any Cause

- It has got features of a renal mass.

Left-Sided Adrenal Mass

- It does *not* move with respiration.
- It is not mobile.
- It is a deeply placed mass.
- Often, it crosses the midline.
- It is resonant on percussion.
- It mimics a kidney mass.

Mass Arising from the Tail of the Pancreas

- Clinical features are same as other pancreatic masses.
- Causes are pseudo-cyst in tail of the pancreas and cyst-adenomas.

Palpable Kidney Mass

Characteristics

- There is fullness in the loin, which is better observed in the sitting position.
- The mass moves with respiration.
- It is vertically placed.
- It is bimanually palpable.
- It is ballotable.
- The renal angle is dull on percussion (normally it is resonant due to colon).
- There is a band of resonance in front due to reflected colon.
- It does *not* cross the midline.

Causes of Enlarged Kidney

1. *Hydronephrosis*
 - It is smooth, soft, and lobulated.
 - It is a non-tender mass
 - It is non-mobile.
2. *Pyonephrosis*
 - History of throbbing pain in the loin, pyuria and fever with chills.
 - It is smooth, soft and tender kidney mass
 - It is non-mobile.
3. *Polycystic kidney*
 - History of loin pain and hematuria.
 - Hypertension, anemia and features of renal failure.
 - Usually bilateral, but one side can present earlier than the other.
 - Lobulated smooth surface.
4. *Renal cell carcinoma*
 - History of a mass in the loin, hematuria, fever and dull pain.
 - The mass is nodular and hard.
 - It does *not* cross the midline.
 - Initially the mass is mobile; eventually it infiltrates gets fixed and becomes non-mobile.

Mass from the Ascending Colon on Right Side or Descending Colon on Left Side

- History of altered bowel habits with decreased appetite and weight.
- The mass is nodular, hard, does not move with respiration and is *not* ballotable.
- It is resonant or there is impaired resonance on percussion.
- Renal angle is resonant.
- Proximal dilated bowel may be palpable.

Adrenal Mass

- The mass is nodular and hard.
- It does *not* move with respiration.
- It is not mobile and often crosses the midline.
- It is felt on deep palpation.
- It is resonant in front.
- It is *not* ballotable.

Retro-peritoneal Tumors and Cysts

- *Retro-peritoneal tumors* are not mobile and do not fall forward in knee-elbow position. They are deeply placed mass which are usually smooth and hard. They may be retro-peritoneal sarcomas or teratomas or lymph node (LN) mass.
- *Retro-peritoneal cysts* are smooth, soft, with same features as retro-peritoneal tumors

N.B. Cystic Lesions in the Abdomen

- Mucocele/empyema of the GB
- Pseudo-cyst of the pancreas
- Hydatid cyst of the liver
- Congenital non-parasitic cyst of the liver
- Hydronephrosis
- Mesenteric cyst
- Ovarian cyst
- Omental cyst
- Aneurysm
- Retro-peritoneal cyst
- Cyst-adeno-carcinoma of the ovary
- Loculated ascites

MASS IN THE UMBILICAL REGION

Common Causes

- Mesenteric cyst
- Omental cyst
- Ovarian cyst (pedunculated)
- Small bowel tumors
- Extension of masses from other region
- Transverse colon mass
- Mass in the body of pancreas
- Mesentery mass
- LN mass: secondaries (primary from GIT, testis, ovary, melanoma), lymphoma, TB
- Retro-peritoneal tumor

Mesenteric Cyst

- *Tillaux Triad*
 1. Soft intra-abdominal umbilical mass.
 2. Mobile in the direction perpendicular to the attachment of the mesentery.
 3. Resonant mass.
- May precipitate intestinal obstruction, volvulus.

Omental Cyst

- It is smooth, soft and non-tender.
- It moves with respiration.
- It is mobile in all directions.
- It is dull on percussion.

Small Bowel Swellings

1. Small bowel lymphomas.
2. Small bowel carcinomas.
3. Intussusception.
 - The Mass in umbilical, usually towards left and above the umbilicus.
 - Occasionally, towards the right side.
 - Mass is intra-abdominal, *sausage-shaped*, with concavity towards umbilicus, well-defined, smooth, firm and mobile.
 - Mass does *not* move with respiration.
 - Mass contracts under palpating fingers.
 - Often, the mass disappears and reappears.
 - It is resonant or there is impaired resonance on percussion.
 - “Red currant jelly” stool with features of intestinal obstruction may be present.

MASS IN THE RIGHT ILIAC FOSSA (RIF)

Common Causes

- Appendicular mass or abscess
- Carcinoma cecum
- Ileo-cecal tuberculosis (TB)
- Ameboma
- Psoas abscess
- Lymph node mass either mesenteric or external Iliac LNs
- Bony swellings
- Ectopic kidney
- Undescended testis (Abdominal)
- Actinomycosis
- Crohn’s disease
- Iliac artery aneurysm
- Gynecological: ovarian cyst, tubo-ovarian mass, uterine mass (pedunculated fibroid)

Appendicular Mass

- It is *smooth, firm, tender mass* in the right iliac fossa.
- It is not mobile and does not move with respiration.
- It is *resonant on percussion*.
- It is a well-localized mass with distinct borders

Appendicular Abscess

- It is *smooth, soft, tender and dull mass* in the right iliac fossa
- It has indistinct borders.

Carcinoma of the Cecum

- It is a nodular, hard mass in the RIF, and does *not* move with respiration.
- It is mobile but mobility may be restricted once it gets adherent to psoas muscle.
- Mass is resonant or there is impaired resonance on percussion.
- Often features of intestinal obstruction may be present.

Ileo-cecal Tuberculosis (TB)

- Mass in the right iliac fossa, which is smooth, hard, resonant and non-tender.
- It does not move with respiration and has a restricted mobility.
- The cecum may be *pulled-the up* to the lumbar region due to fibrosis.

Ameboma

- History of dysentery with pain in the RIF.
- Smooth, hard, well-defined mass in the RIF, non-mobile.
- It may or may not be tender.

Psoas Abscess

- It is a localized, smooth, soft, non-mobile mass in the RIF.
- *Psoas spasm* (flexion of the hip joint) is typical.
- Spine may show *tenderness and para-spinal spasm* (spinal movements are restricted)

5. MASS IN THE LEFT ILIAC FOSSA (LIF)

Causes

1. Carcinoma sigmoid or descending colon
2. Bony masses
3. Ovarian/uterine masses
4. Psoas abscess
5. Ectopic kidney
6. Lymph node mass
7. Undescended testis

MASS IN THE HYPOGASTRIUM

Bladder Mass

- It is in the midline.
- It is dull on percussion.
- The lower border is not felt.
- It can be mobile in a horizontal direction.
- The mass reduces in size after emptying the bladder.
- It can be felt on per-rectal examination.
- It is either carcinoma bladder (common) or leiomyoma or sarcoma bladder.

Uterine Mass

- It is a midline mass, which is smooth, hard.
- The lower border, which extends into the pelvis, is not felt.
- It is felt on per-vaginal (PV) examination.

Ovarian Mass

- It is a pelvic soft tissue mass.
- As in all lower abdomen masses PR and/or PV examination is a must.

INVESTIGATIONS OF AN ABDOMINAL MASS

Laboratory Test

- Hematocrit value (HCV)
- Liver function tests (LFTs)
- Renal function tests
- Stool/urine analysis.

Imaging Studies

- US abdomen.
- Barium studies: Barium meal - Barium enema - Barium meal - Follow through.
- CT scan
- MRI - MRCP
- Endo-sonography.

Endoscopic/Laparoscopic Procedures

- Gastroscopy
- Colonoscopy
- ERCP.
- Diagnostic laparoscopy.

Tapping

- Ascitic tap

Biopsy

- US-guided biopsy
- CT-guided biopsy.

Renal Investigations

- Laboratory: Serum urea and creatinine
- Imaging: IVU - RGP - isotope renogram
- Endoscopy: Cystoscopy.

Exploration

- Exploratory laparotomy.

IMPORTANT CLINICAL NOTES

- A hard mass in the abdomen is commonly malignant
- A firm mass may be tuberculous, lymphoma or many benign conditions
- A soft mass is hydronephrosis, pseudo-cyst, mesenteric cyst, omental cyst, or loculated ascites
- In TB abdomen, the mass may be doughy in consistency due to thickened parietal peritoneum or omentum
- It is difficult to elicit fluctuation in the abdominal mass as mass cannot be fixed properly
- Plane of the swelling should be checked by leg raising/head-raising test / Valsalva manoeuvre / knee-elbow test
- Bimanual palpation, ballotability, renal angle inspection, palpation and percussion should be done in case of a renal mass
- Intrinsic mobility should be checked. Different mobilities / movements are—gallbladder shows side to side; stomach lateral; ovarian mass—all over; mesenteric cyst—right angle to line of mesentery; transverse colon mass—vertical; small bowel mass all over; appendicular mass/pancreatic mass/ nodal mass (para-aortic)/retroperitoneal mass do not show any mobility; cystadenocarcinoma of pancreas may show false mobility (*tree top mobility*)
- Percussion is very important method of examination to find out the anatomical plane of the mass. Mass in front of the bowel like liver/spleen/GB/parietal mass is dull on percussion. Mass from the bowel is resonant on percussion e.g. stomach, small bowel and colon. Retro-peritoneal masses like cyst, sarcoma, nodal mass, aneurysm, pancreatic mass are resonant on percussion
- Succussion splash and auscultation-percussion tests are done for gastric outlet obstruction in pyloric stenosis
- Digital examination of rectum (PR) and left supraclavicular examination for nodes is a must. PR is done to see Blumer shelf secondaries in the recto-vesical pouch. Per-vaginal examination should be done in pelvic masses in females
- The urinary bladder should be emptied while examining a pelvic mass
- Bimanual examination is done often under general anesthesia in case of a pelvic mass

- Auscultation for bruit depends on condition and location of the mass (over epigastrium, over liver)
- Other relevant systemic examination is a must in examination of mass abdomen — respiratory system, skeletal and neurological systems
- *Masses that may appear and disappear*
 1. Pseudocyst of pancreas (communicating)
 2. Hydronephrosis (intermittent)
 3. Choledochal cyst
 4. Intussusception